
From Lyapunov Analysis to Algorithm Design in two-sided PL Minimax Optimization

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Abstract

We derive algorithms for smooth nonconvex non-concave minimax optimization and establish linear convergence rates for problems that satisfy the two-sided Polyak-Lojasiewicz (PL) inequality. At the core of our approach is the observation that Lyapunov functions can be used not only to certify convergence a posteriori, but also to design algorithms. By replacing an idealized, intractable Lyapunov function with a computable surrogate based on gradient information, we derive TALDA (Tri-Action Lyapunov Descent Ascent), a single-loop algorithm that enforces Lyapunov descent by construction. TALDA guarantees linear convergence under the two-sided PL condition, with a rate that depends explicitly on the cross-smoothness constant. This recovers existing worst-case guarantees while yielding sharper convergence rates in weakly coupled min-max problems.

1. Introduction

Lyapunov functions are a central tool for analyzing the stability of a dynamical system. For instance, a Lyapunov function associates each state with a scalar “energy” whose monotonic decrease along trajectories provides a certificate of stability (Lyapunov, 1992). As a result, Lyapunov functions have also become a cornerstone in optimization: they offer a unifying conceptual framework (Chen & Luo, 2021; Wilson et al., 2021; Laborde & Oberman, 2020; Diakonikolas & Wang, 2022; Er et al., 2025) and are widely used to establish convergence guarantees (Wilson, 2018; Bansal & Gupta, 2017; Muehlebach & Jordan, 2021).

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In the context of minimization problems, Diakonikolas & Orecchia (2019) unify the derivation of first-order methods decreasing an approximate duality gap. Du (2022) design algorithms for submodular maximization via a two-stage Lyapunov-based approach. Gramlich et al. (2020) formulate Lyapunov conditions as linear matrix inequalities to systematically design accelerated methods for minimizing functions beyond the strongly convex regime, as well as for a restricted class of saddle point problems. More recently, Upadhyaya et al. (2025) uses Lyapunov functions to design novel extragradient-type algorithms for monotone variational inequalities.

Despite these advances, existing Lyapunov-based algorithm design frameworks have largely focused on minimization problems, convex-concave saddle point problems, or monotone variational inequalities. In contrast, much less is understood about how Lyapunov functions can be leveraged for algorithm design in min-max problems of the form

$$\min_{\mathbf{x}} \max_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}), \quad f : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}. \quad (1)$$

A wide variety of machine learning applications, including generative adversarial networks (GANs) (Goodfellow et al., 2014; Arjovsky et al., 2017), adversarial machine learning (Madry et al., 2018; Sinha et al., 2018), robust optimization (Duchi & Namkoong, 2019), and many others, can be formulated as in Equation (1). Unlike pure minimization problems, min-max optimization is complicated by the presence of two competing objectives, often leading to oscillatory or divergent dynamics (Hemmat et al., 2023; Gidel et al., 2019b; Mescheder et al., 2017). Since Lyapunov functions are often used to design stabilizing controllers for dynamical systems (Khalil, 2008), this raises the central question that motivates our work:

Can we find a principled way to design algorithms for min-max problems using Lyapunov functions?

Contributions. We investigate this question for a subclass of smooth nonconvex-nonconcave min-max problems that satisfy the two-sided PL inequality (Yang et al., 2020).

- We construct a computable Lyapunov surrogate that upper bounds the primal–dual gap (Section 3). We

Algorithm 1 : TALDA (Tri-Action Lyapunov Descent Ascent)

1: Main loop 2: Input: oracle of $f, \epsilon, (\mathbf{x}_0, \mathbf{y}_0), \mu_f, L_f, L_{\mathbf{x}\mathbf{y}}, \delta$. 3: Init: $k \leftarrow 0; (\hat{\mathbf{x}}_0, \hat{\mathbf{y}}_0) \leftarrow (\mathbf{x}_0, \mathbf{y}_0)$. 4: $V_0 \leftarrow \frac{1}{2\mu_f} (\ \nabla_{\mathbf{x}} f(\mathbf{x}_0, \mathbf{y}_0)\ ^2 + \ \nabla_{\mathbf{y}} f(\mathbf{x}_0, \mathbf{y}_0)\ ^2)$. 5: while $V(\mathbf{x}_k, \mathbf{y}_k, \hat{\mathbf{x}}_k, \hat{\mathbf{y}}_k) \geq \epsilon$ do 6: $\hat{\mathbf{x}}_{k+1} \leftarrow \hat{\mathbf{x}}_k - \frac{1}{L_f} \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}_k, \mathbf{y}_k)$. 7: $\hat{\mathbf{y}}_{k+1} \leftarrow \hat{\mathbf{y}}_k + \frac{1}{L_f} \nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}_k)$. 8: $(\mathbf{x}_{k+1}, \mathbf{y}_{k+1}, \hat{\mathbf{x}}_{k+1}, \hat{\mathbf{y}}_{k+1}, V_{k+1})$ 9: $\leftarrow \text{LyapUpdate}(\mathbf{x}_k, \mathbf{y}_k, \hat{\mathbf{x}}_{k+1}, \hat{\mathbf{y}}_{k+1}, V_k)$. 10: $k \leftarrow k + 1$. 11: end while 12: Output: $(\mathbf{x}_{\text{out}}, \mathbf{y}_{\text{out}}) \leftarrow (\mathbf{x}_k, \mathbf{y}_k)$.	1: Subroutine: LyapUpdate 2: Params: f, r from Eq. 22, η from Eq. 16. 3: Input: $\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, V_{\text{old}}$. 4: $\mathbf{x}^{\text{grad}} \leftarrow \mathbf{x} - \eta \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})$. 5: $\mathbf{y}^{\text{grad}} \leftarrow \mathbf{y} + \eta \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y})$. 6: $V_1 \leftarrow V(\mathbf{x}^{\text{grad}}, \mathbf{y}^{\text{grad}}, \hat{\mathbf{x}}, \hat{\mathbf{y}})$. 7: $V_2 \leftarrow V(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \mathbf{x}, \mathbf{y})$. 8: If $V_1 \leq \min\{V_2, (1-r)V_{\text{old}}\}$ (Grad is best) 9: Return $(\mathbf{x}^{\text{grad}}, \mathbf{y}^{\text{grad}}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, V_1)$ 10: If $V_2 \leq \min\{V_1, (1-r)V_{\text{old}}\}$ (Swap is best) 11: Return $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \mathbf{x}, \mathbf{y}, V_2)$ 12: Default: (Continue approximation) 13: Return $(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, V_{\text{old}})$
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further derive constructive descent conditions that enable algorithmic updates without exact best-response oracles (Section 4).

- The descent conditions give rise to **TALDA** (Tri-Action Lyapunov Descent Ascent; Algorithm 1), a single-loop, decision-based algorithm that guarantees Lyapunov contractions and relies entirely on first-order oracles.
- We establish linear convergence of **TALDA** under the two-sided PL condition, with iteration complexity $\tilde{O}\left((\kappa_f^2 + \kappa_f \kappa_{\mathbf{x}\mathbf{y}}) \log^2(1/\epsilon)\right)$, yielding optimal rates in weakly coupled regimes. To the best of our knowledge, this is the first analysis providing sharp convergence guarantees in this setting (Theorem 1).

Section 5 further shows that the analysis is tight by recovering known convergence guarantees in idealized settings with exact best-response oracles and in uncoupled min-max problems. Finally, Section 6 evaluates the proposed methods empirically and demonstrates their effectiveness relative to standard baselines.

1.1. Preliminaries

Bold lower-case letters are used to denote vectors, like $\mathbf{x}, \mathbf{y}, \mathbf{z}$. Let $f(\cdot, \cdot) : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}$ be a function of two variables, $\nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})$ denotes the partial gradient of f with respect to $\mathbf{x}$, similarly, $\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})$ denotes the partial gradient of f with respect to $\mathbf{y}$. For a vector $\mathbf{x}$, $\|\mathbf{x}\|$ denotes the l_2 -norm of $\mathbf{x}$. $O(\cdot), \Omega(\cdot)$ are used to hide absolute constants that do not depend on the problem parameters. $\tilde{O}(\cdot), \tilde{\Omega}(\cdot)$ are used to hide absolute constants and log factors.

We say that a differentiable function $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ is smooth

and satisfies the PL inequality if, respectively,

$$\begin{aligned} |\phi(\mathbf{x}_2) - \phi(\mathbf{x}_1) - \nabla \phi(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1)| &\leq \frac{L}{2} \|\mathbf{x}_2 - \mathbf{x}_1\|^2, \\ \phi(\mathbf{x}) - \min_{\mathbf{x}} \phi(\mathbf{x}) &\leq \frac{1}{2\mu} \|\nabla \phi(\mathbf{x})\|^2. \end{aligned} \quad (2)$$

For instance, functions that are strongly convex, i.e.,

$$\phi(\mathbf{x}_2) - \phi(\mathbf{x}_1) - \nabla \phi(\mathbf{x}_1)^T (\mathbf{x}_2 - \mathbf{x}_1) \geq \frac{\mu}{2} \|\mathbf{x}_2 - \mathbf{x}_1\|^2, \quad (3)$$

also satisfy the PL inequality with the same constant. The converse is not true, as PL functions can be nonconvex.

We also note the important distinction between a Lyapunov and a Lyapunov candidate function:

Definition 1 (Lyapunov candidate function). Let $\mathcal{S} \subset \mathbb{R}^{d_x+d_y}$ denote the set of equilibrium points of Equation (1). A function $V : \mathbb{R}^{d_x+d_y} \rightarrow \mathbb{R}_{\geq 0}$ is a *Lyapunov candidate function* if $V(\xi) > 0$ for all $\xi \notin \mathcal{S}$ and $V(\xi) = 0$ if and only if $\xi \in \mathcal{S}$.

Definition 2 (Lyapunov function). A Lyapunov candidate function V is a *Lyapunov function* for a discrete-time system $\{\xi_k\}_{k \geq 0}$ if $V(\xi_{k+1}) < V(\xi_k)$ for all $k \geq 0$.

1.2. Related Work

Several works on nonconvex-nonconcave min-max optimization leverage the PL condition to obtain convergence rates, as it provides a useful structure to the problem. Early efforts, such as Nouiehed et al. (2019), investigated the one-sided PL condition (non-convex but concave in the dual variable), establishing convergence guarantees for multi-step Gradient Descent Ascent (GDA). Yang et al. (2020) expanded the result to the two-sided setting and established a critical structural property: under the two-sided PL condition, every stationary point is a global Nash equilibrium. This result implies that despite potential non-convexity, local

search methods can find global optima. Furthermore, Yang et al. (2020) and Chen et al. (2022) proved that Alternating Gradient Descent Ascent (AGDA) converges linearly at the rate of $O(\kappa_x \kappa_y^2 \log(\epsilon^{-1}))$ in terms of gradient evaluations with condition numbers $\kappa_x = L_x/\mu_x$ and $\kappa_y = L_y/\mu_y$, challenging the prior intuition that linear rates require strong convexity/concavity.

Beyond GDA-based algorithms, extragradient and related optimistic methods are often used to solve min-max problems. However, to the best of our knowledge, there are no existing results establishing linear convergence of extragradient methods under a two-sided PL condition without strong monotonicity. The primary reason is that in the unconstrained settings, extra-gradient and related optimistic methods are typically analyzed via contraction arguments on the distance to a solution, which rely on strong monotonicity of the underlying operator. In smooth nonconvex-nonconcave problems, however, such contraction properties generally do not hold, and extra-gradient methods no longer admit last-iterate guarantees without additional assumptions. For instance, recent works such as Liu et al. (2023) impose bounded-domain assumptions to obtain convergence guarantees on averaged iterates of the extra-gradient algorithm. As a result, the literature on smooth, unconstrained two-sided PL min-max problems has predominantly centered on GDA-based algorithms.

While our work focuses on the deterministic two-sided PL setting, many existing works under the two-sided PL setting investigate the stochastic case. For example, Yang et al. (2020) provides convergence rates for stochastic AGDA and a variance-reduced algorithm SVRG, followed by Chen et al. (2022), who propose SPIDER-GDA as an alternative to the variance-reduced SVRG-AGDA, deriving an improved convergence rate in the stochastic case. Das et al. (2022) and Cho & Yun (2023) provided a comprehensive analysis of Stochastic Gradient Descent Ascent (SGDA), specifically investigating the impact of data shuffling (sampling without replacement). Importantly, when applied to the deterministic setting, stochastic AGDA, SVRG, SPIDER-GDA, and SGDA all converge with a similar linear rate of $O(\kappa_x \kappa_y^2 \log(\epsilon^{-1}))$ gradient evaluations. Furthermore, AccSPIDER-GDA (Chen et al., 2022) introduces a Catalyst-style acceleration framework that improves the rate of SPIDER-GDA to a deterministic accelerated rate of $O(\kappa_x \kappa_y \log(\frac{\kappa_y}{\epsilon}) \log(\epsilon^{-1}))$.

However, existing methods typically derive rates that do not explicitly capture the coupling between the two players. In contrast, our algorithm has an iteration complexity of $O((\kappa_f^2 + \kappa_f \kappa_{xy}^2) \log^2(\epsilon^{-1}))$ where $\kappa_f = \max\{\kappa_x, \kappa_y\}$, and $\kappa_{xy} = L_{xy}/\mu_f$. Accounting for the interaction between the min and the max player through κ_{xy}^2 instead of κ_y^2 results in an improved convergence rate, most notably when

the interaction between the players is weak ($\kappa_{xy} < \sqrt{\kappa_f}$).

2. Assumptions

Assumption 1 (S2SPL — Smooth two-sided PL). For all $\mathbf{y}$ (resp. $\mathbf{x}$), the function $f(\cdot, \mathbf{y})$ (resp. $-f(\mathbf{x}, \cdot)$) is L_f -smooth and satisfies the PL inequality with constant μ_f . Moreover, f is jointly smooth in the sense that

$$\begin{aligned} \|\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{y}} f(\mathbf{x}', \mathbf{y})\| &\leq L_{xy} \|\mathbf{x} - \mathbf{x}'\|, \\ \|\nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}')\| &\leq L_{xy} \|\mathbf{y} - \mathbf{y}'\|. \end{aligned} \quad (4)$$

Remark. More generally, one could distinguish separate smoothness constants L_x and L_y for $f(\cdot, \mathbf{y})$ and $-f(\mathbf{x}, \cdot)$, respectively, as well as separate PL constants μ_x and μ_y . For clarity of exposition, we use L_f and μ_f throughout.

S2SPL includes strongly convex-strongly concave objectives; PL-strongly concave functions; and a wide range of nonconvex-nonconcave objectives. Some examples of applications include robust least squares, generative adversarial imitation learning for linear-quadratic regulator dynamics (Cai et al., 2019), and zero-sum linear-quadratic games (Zhang et al., 2019), and arise more broadly in adversarial learning (Du et al., 2019), robust phase retrieval (Sun et al., 2018; Zhou et al., 2016), and robust control (Fazel et al., 2018). Additionally, recent work by Patel & Vlatakis-Gkaragkounis (2026) shows that under overparameterization, hidden convex-concave structure reduces to a two-sided PL condition with high probability, suggesting that problems such as GANs, robust adversarial reinforcement learning (RARL), and adversarial example generation (Patel & Vlatakis-Gkaragkounis, 2026, Appendix B) can satisfy S2PL.

This class of functions enjoys the following property.

Lemma 1 (Yang et al. (2020, Lem 2.1)). *Under A. 1, saddle points, global minimax, and stationary points coincide.*

Assumption 2 (Existence of saddle points). For all $\mathbf{y}$ (resp. $\mathbf{x}$), there exist at least a solution to the problem $\min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y})$ (resp. $\max_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})$)

Under A. 2, the set of minimizers and the set of maximizers are nonempty. We fix arbitrary selections $\mathcal{X}(\cdot)$ and $\mathcal{Y}(\cdot)$ s.t.

$$\mathcal{X}(\mathbf{y}) \in \arg \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}), \quad \mathcal{Y}(\mathbf{x}) \in \arg \max_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}). \quad (5)$$

For convenience, we also define the two condition numbers that are used throughout the article:

$$\kappa_f := \frac{L_f}{\mu_f} \geq 1, \quad \kappa_{xy} := \frac{L_{xy}}{\mu_f} \geq 0. \quad (6)$$

For some algorithms (not TALDA), we need the following.

Assumption 3 (Cross second-order Hessians). The mixed Hessians $\nabla_{xy}^2 f(\mathbf{x}, \mathbf{y})$ and $\nabla_{yx}^2 f(\mathbf{x}, \mathbf{y})$ exist $\forall (\mathbf{x}, \mathbf{y})$.

3. Computable Lyapunov Function

The following section presents our methodology for Lyapunov-based algorithm design. The central idea is to start from a Lyapunov candidate function in the sense of Def. 1, and to enforce its decrease along the iterates resulting in a Lyapunov function as defined in Def. 2.

3.1. Idealized Lyapunov Function

We define the *idealized Lyapunov function* as

$$V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) := \max_{\mathbf{y}'} f(\mathbf{x}, \mathbf{y}') - \min_{\mathbf{x}'} f(\mathbf{x}', \mathbf{y}), \quad (7)$$

which measures primal-dual suboptimality with respect to the exact best responses. By Lemma 1, $V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) = 0$ if and only if $(\mathbf{x}, \mathbf{y})$ is a saddle point.

As a consequence, the original min-max problem in Equation (1) can be equivalently viewed as the **minimization** of the nonnegative function V^{Exact} over $(\mathbf{x}, \mathbf{y})$. Hence, by applying gradient descent directly to V^{Exact} , we converge toward the solution of Equation (1).

It is known that $V^{\text{Exact}}(\xi)$ is a smooth function in $\xi = (\mathbf{x}, \mathbf{y})$ satisfying the PL inequality, with the following constants: $L_{V^{\text{Exact}}} = L_f + L_{\mathbf{xy}}^2/\mu_f$ and $\mu_{V^{\text{Exact}}} = \mu_f$ (Nouiehed et al., 2019). Therefore, gradient descent on V^{Exact} with step-size $\frac{1}{L_{V^{\text{Exact}}}}$ converges linearly (see Theorem 2),

$$V^{\text{Exact}}(\xi_k) \leq (1 - (\kappa_f + \kappa_{\mathbf{xy}}^2)^{-1})^k V^{\text{Exact}}(\xi_0). \quad (8)$$

Despite its appealing theoretical properties, exact computation of the Lyapunov function V^{Exact} is intractable, as it depends on the exact best-response mappings $\mathcal{X}(\mathbf{y})$ and $\mathcal{Y}(\mathbf{x})$. Nevertheless, V^{Exact} provides a natural performance baseline, characterizing a linear convergence rate under the S2SPL structure (see A. 1) and serving as a reference for comparing algorithms.

3.2. Computable Lyapunov Functions

A key observation is that, under A. 1, suboptimality can be bounded by gradient norms through the PL inequality (2). In particular, for any fixed $\mathbf{y}$ and $\mathbf{x}$, and for all $\tilde{\mathbf{x}}, \tilde{\mathbf{y}}$,

$$\begin{aligned} f(\tilde{\mathbf{x}}, \mathbf{y}) - f(\mathcal{X}(\mathbf{y}), \mathbf{y}) &\leq \frac{1}{2\mu_f} \|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}, \mathbf{y})\|^2, \\ f(\mathbf{x}, \mathcal{Y}(\mathbf{x})) - f(\mathbf{x}, \tilde{\mathbf{y}}) &\leq \frac{1}{2\mu_f} \|\nabla_{\mathbf{y}} f(\mathbf{x}, \tilde{\mathbf{y}})\|^2. \end{aligned} \quad (9)$$

Motivated by Equation (9), we replace the intractable best-response terms in V^{Exact} by their PL upper bounds. This leads to a *computable* surrogate Lyapunov function $V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ that depends only on function values and gra-

dients at $(\mathbf{x}, \mathbf{y})$ and auxiliary points $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$,

$$\begin{aligned} V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) &= f(\mathbf{x}, \tilde{\mathbf{y}}) - f(\tilde{\mathbf{x}}, \mathbf{y}) \\ &\quad + \frac{1}{2\mu_f} (\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}, \mathbf{y})\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}, \tilde{\mathbf{y}})\|^2). \end{aligned} \quad (10)$$

By construction, $V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ provides a natural computable upper bound on $V^{\text{Exact}}(\mathbf{x}, \mathbf{y})$ that coincides at saddle points, and is therefore a Lyapunov candidate function.

Lemma 2. *If f satisfies A. 1 and 2, the Lyapunov functions in Equations (7) and (10) satisfy, $\forall (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$,*

$$V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) \leq V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \quad \forall (\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}. \quad (11)$$

In addition, $V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) = \min_{\tilde{\mathbf{x}}, \tilde{\mathbf{y}}} V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}})$.

In the rest of the paper, when unambiguous, we will write $\xi = (\mathbf{x}, \mathbf{y})$, $\tilde{\xi} = (\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$, and $\hat{\xi} = (\hat{\mathbf{x}}, \hat{\mathbf{y}})$ for brevity.

4. Designing Algorithms Using Lyapunov Functions: A Constructive Approach

It remains to design a method that enforces a strict decrease of V in Equation (10) until we reach an accuracy ε ,

$$V_T \leq \varepsilon, \quad V_{k+1} \leq V_k, \quad V_k := V(\xi_k, \tilde{\xi}_k).$$

By Lemma 2, minimizing V is sufficient to recover a saddle point of Equation (1), as V provides a global upper bound on V^{Exact} and shares the same minimizer. Unlike V^{Exact} , V is generally nonsmooth, nonconvex, and does not satisfy the PL inequality. This prevents a direct application of gradient descent and necessitates more careful algorithmic design.

Direct minimization using GD. Under A. 3, the direct minimization of V using gradient descent leads to update rules involving Hessian–vector products. These terms are computationally tractable, as they do not require forming or inverting Hessian matrices (see Lemma 7 in Appendix B), and can be interpreted as look-ahead corrections. The resulting Algorithm 3 is closely related to LCGD and first-order LOLA (Schäfer & Anandkumar, 2019; Foerster et al., 2017), in the difference that it operates on four variables and evaluates gradients at distinct points. Although this approach guarantees strict Lyapunov descent, deriving explicit convergence rates is nontrivial, as V is generally nonconvex and may be nonsmooth under our assumptions. This observation motivates the alternative construction developed next.

4.1. Main Idea

Our approach manipulates three types of iterates: The primary $\xi_k = (\mathbf{x}_k, \mathbf{y}_k)$ is the quantity whose convergence is guaranteed and returned by the algorithm. The auxiliary iterate $\tilde{\xi}_k$ is used to certify contraction of the Lyapunov

function across iterations. Finally, $\hat{\xi}_k$ is an approximate best-response point, whose role is to reduce the gap between the computable Lyapunov surrogate V and V^{Exact} .

The minimization of V proceeds in two stages. First, we compute approximate best-response points $\hat{\xi}_{k+1}$ such that

$$V(\xi_k, \hat{\xi}_{k+1}) \approx \min_{\hat{\xi}} V(\xi_k, \hat{\xi}) \stackrel{(\text{Lemma 2})}{=} V^{\text{exact}}(\xi_k).$$

Second, based on this approximation, we update the iterates $(\xi_{k+1}, \tilde{\xi}_{k+1})$ to enforce a contraction:

$$V(\xi_{k+1}, \tilde{\xi}_{k+1}) \leq (1-r)V(\xi_k, \tilde{\xi}_k), \quad (12)$$

where r will be determined later (see Equation (22)). The underlying intuition is to approximate a gradient descent step on the idealized Lyapunov $V^{\text{Exact}}(\mathbf{x}, \mathbf{y})$ by selecting $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ sufficiently close to the exact best responses $\mathcal{X}(\mathbf{y}_k)$ and $\mathcal{Y}(\mathbf{x}_k)$, thereby retaining the favorable convergence properties of the idealized dynamics.

This strategy raises three challenges: V is nonconvex and does not satisfy a PL condition, complicating the computation of accurate inner approximations; updates of the outer variables may invalidate previously computed best responses; and the lack of smoothness and PL structure in V makes linear convergence difficult to establish.

The first difficulty is, however, only superficial. Inspecting V in Equation (10) reveals that any $\mathcal{X}(\mathbf{y}_k)$ and $\mathcal{Y}(\mathbf{x}_k)$ jointly minimize $V(\mathbf{x}_k, \mathbf{y}_k, \cdot, \cdot)$. Under the S2SPL assumption, these approximations can be efficiently approximated using gradient descent/ascent on f .

The remaining two challenges are more subtle. Nevertheless, if $\hat{\mathbf{x}}_{k+1}, \hat{\mathbf{y}}_{k+1}$ are sufficiently accurate, a natural algorithmic structure emerges by updating $(\mathbf{x}_k, \mathbf{y}_k)$ via a gradient step on the surrogate objective $V(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}_{k+1}, \hat{\mathbf{y}}_{k+1})$.

4.2. Computing the Approximation

We assume for now that the approximations $\hat{\mathbf{x}}_{k+1} \approx \mathcal{X}(\mathbf{y}_k)$ and $\hat{\mathbf{y}}_{k+1} \approx \mathcal{Y}(\mathbf{x}_k)$ can be computed efficiently. We require that, for $r \in [0, 1]$ (that will be later the convergence rate), the approximation $\hat{\xi}_{k+1} = (\hat{\mathbf{x}}_{k+1}, \hat{\mathbf{y}}_{k+1})$ satisfies

$$\begin{aligned} V(\xi_k, \hat{\xi}_{k+1}) - \min_{\hat{\xi}} V(\xi_k, \hat{\xi}) \\ \leq (1-r)(V(\xi_k, \tilde{\xi}_k) - \min_{\hat{\xi}} V(\xi_k, \hat{\xi})), \end{aligned} \quad (13)$$

or equivalently (see Lemma 2),

$$V(\xi_k, \hat{\xi}_{k+1}) \leq (1-r)V(\xi_k, \tilde{\xi}_k) + rV^{\text{Exact}}(\xi_k). \quad (14)$$

In the worst case, gradient descent can achieve this condition after $T_{\text{inner}} = \mathcal{O}(\kappa_f \log(\varepsilon^{-1}))$ iterations (see Corollary 1).

The right-hand side of Equation (14) reveals an inherent trade-off. While a larger contraction rate r accelerates the

decrease of $V(\xi_k, \hat{\xi}_{k+1})$, it also places greater weight on the perturbation term $V^{\text{Exact}}(\xi_k)$, which cannot be directly controlled. This term must therefore be compensated by the subsequent update of $(\mathbf{x}_{k+1}, \mathbf{y}_{k+1})$; if r is chosen too large, this compensation may be insufficient, preventing an overall decrease of the Lyapunov function.

4.3. Effect of a Gradient Step

We analyze the effect of updating the iterate ξ_k with a gradient descent step on f after computing an auxiliary point $\hat{\xi}_{k+1}$ by approximately minimizing the Lyapunov surrogate (see Equation (13)). The goal of this section is to characterize the resulting change in the Lyapunov value.

The next lemma provides a sufficient condition on the approximation quality of $\hat{\xi}_{k+1}$, expressed solely in terms of gradient information of f , under which a single gradient step on f guarantees a decrease of the Lyapunov function.

Lemma 3. *Under A. 1, if $\hat{\xi} = (\hat{\mathbf{x}}, \hat{\mathbf{y}})$ satisfies for $\delta \in [0, 1]$*

$$\left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \\ \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \end{bmatrix} \right\| \leq \frac{\delta}{\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \\ \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \end{bmatrix} \right\|, \quad (15)$$

then the gradient step $\xi^{\text{grad}} = (\mathbf{x}^{\text{grad}}, \mathbf{y}^{\text{grad}})$ on f defined by

$$\begin{aligned} \mathbf{x}^{\text{grad}} &= \mathbf{x} - \eta \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \mathbf{y}^{\text{grad}} &= \mathbf{y} + \eta \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{aligned}, \quad \eta = \frac{1-\delta}{L_f + L_{\mathbf{xy}} \kappa_{\mathbf{xy}}} \quad (16)$$

guarantees that the Lyapunov decrease satisfies

$$V(\xi, \hat{\xi}) - V(\xi^{\text{grad}}, \hat{\xi}) \geq \frac{(1-\delta)^2/2}{L_f + L_{\mathbf{xy}} \kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2.$$

It remains to know if Equation (15) can be achieved. Crucially, if the gradient norm on the left-hand side of Equation (15) becomes sufficiently small, then $V_k \leq \varepsilon$, and the algorithm can be terminated. This is discussed in more detail in the proof of Theorem 1, see Equation (34).

Interpretation of δ . The user-defined parameter $\delta \in [0, 1]$ quantifies how well $\hat{\xi}$ approximates a best-response pair. Condition (15) requires the “cross” gradient components $\nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}})$ and $\nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y})$ to be small relative to the “direct” gradients that drive descent of the Lyapunov surrogate. Smaller values of δ correspond to more accurate inner approximations, which allow for larger step sizes and yield a stronger guaranteed Lyapunov decrease, but makes the approximation harder to compute.

It remains to analyze if this gradient step is sufficient to compensate the term in Equation (14).

4.4. Decreasing the Lyapunov Function

Assume that the approximation $\hat{\xi}_{k+1}$ satisfies both Equations (14) and (15). We now compare the residual term in

Equation (14) with the decrease induced by the gradient step described in Equation (16). Define

$$\Delta_{k+1} = rV^{\text{Exact}}(\xi_k) - \frac{(1-\delta)^2/2}{L_f + L_{\mathbf{x}\mathbf{y}}\kappa_{\mathbf{x}\mathbf{y}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2, \quad (17)$$

which measures whether the decrease from the outer gradient step compensates the term in Equation (14). The quantity Δ_{k+1} may be either negative or positive.

Case $\Delta_{k+1} \leq 0$. When $\Delta_{k+1} \leq 0$, setting

$$\xi_{k+1} = (\mathbf{x}_{k+1}^{\text{grad}}, \mathbf{y}_{k+1}^{\text{grad}}), \quad \tilde{\xi}_{k+1} = \hat{\xi}_{k+1}, \quad (18)$$

immediately yields a contraction of the Lyapunov function:

$$\begin{aligned} V(\xi_{k+1}, \tilde{\xi}_{k+1}) &\leq V(\xi_k, \hat{\xi}_{k+1}) - \frac{(1-\delta)^2}{2L_f} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 \\ &\leq (1-r)V(\xi_k, \tilde{\xi}_k), \end{aligned} \quad (19)$$

where the first inequality follows from Lemma 3 and the second from Equation (14) together with $\Delta_{k+1} \leq 0$.

Case $\Delta_{k+1} > 0$. When $\Delta_{k+1} > 0$, the gradient step is insufficient to offset the idealized term in Equation (14), and thus cannot guarantee a decrease of the Lyapunov. This situation is not detrimental: combining $\Delta_{k+1} > 0$ with Equation (15) implies that the auxiliary point $\hat{\xi}_{k+1}$ already exhibits a small gradient norm and is therefore closer to a saddle point than ξ_k .

Lemma 4. Assume that $\hat{\xi}_{k+1}$ satisfies Equations (14) and (15) with $\kappa_{\mathbf{x}\mathbf{y}} > 0$ and that $\Delta_{k+1} > 0$. Then

$$\begin{aligned} V(\hat{\xi}_{k+1}, \xi_k) &\leq [r\gamma(\delta)(\kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2) - 1] V(\xi_k, \tilde{\xi}_k), \\ \text{where } \gamma(\delta) &:= \frac{1 + (\delta/\kappa_{\mathbf{x}\mathbf{y}})^2}{(1-\delta)^2}. \end{aligned} \quad (20)$$

Lemma 4 shows that, if r is small, a Lyapunov decrease may be achieved by swapping the roles of the iterates:

$$\xi_{k+1} = \hat{\xi}_{k+1}, \quad \tilde{\xi}_{k+1} = \xi_k. \quad (21)$$

4.5. Rate of Convergence

Since Δ_{k+1} may be positive or negative, the Lyapunov decrease is controlled by either Equation (19) or Equation (20). An optimal contraction rate is obtained by balancing these two bounds, which yields

$$r = \frac{2}{1 + \gamma(\delta)(\kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2)}, \quad \gamma(\delta) = \frac{1 + (\delta/\kappa_{\mathbf{x}\mathbf{y}})^2}{(1-\delta)^2}. \quad (22)$$

In particular, choosing $\delta = \min\{\kappa_{\mathbf{x}\mathbf{y}}, \delta_{\max}\}$ ensures that $\gamma(\delta)$ remains bounded, with $\gamma(\delta) \leq 2/(1-\delta_{\max})^2$.

This rate matches that of gradient descent on the idealized Lyapunov Equation (7), up to the multiplicative factor $\gamma(\delta)$, which captures the effect of inexact inner approximations. Together with Equation (12), this means the Lyapunov function converges linearly toward zero with rate r .

4.6. Single-loop Algorithm

From the previous section, we have seen that once we find a point $\hat{\xi}$ that satisfies Equations (14) and (15), then we are guaranteed that one of the two update in Equation (18) or Equation (21) guarantee a linear decrease in the Lyapunov function. Therefore, instead of having two nested loops – one that first computes $\hat{\xi}_{k+1}$, another that updates ξ_{k+1} and $\tilde{\xi}_{k+1}$ – we can test both updates at every iteration, and proceed if we observe a contraction in the Lyapunov function. In addition, the iterates $\tilde{\xi}_k$ can be simplified, as their only use are to track the Lyapunov values. This is formalized by Algorithm 1.

4.7. Total Number of Iterations

The iteration complexity of Algorithm 1 can be bounded by $O((\kappa_f^2 + \kappa_f\kappa_{\mathbf{x}\mathbf{y}}^2) \log(1/\varepsilon)^2)$. This complexity is determined by two factors: the number of gradient descent steps required to satisfy Equations (14) and (15), and the number of outer iterations needed to reach $V_k \leq \varepsilon$ at the contraction rate given in Equations (12) and (22). The iteration complexity of the former can be bounded by $O(\kappa_f \log(1/\varepsilon))$, while the second is bounded by $O((\kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2) \log(1/\varepsilon))$.

Theorem 1. Let f satisfy A. 1 and 2. After T iterations,

$$T \geq O\left(\kappa_f(\kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2) \log^2(\varepsilon^{-1})\right),$$

the output $\mathbf{x}_T, \mathbf{y}_T$ of Algorithm 1 with $\delta = \min\{\kappa_{\mathbf{x}\mathbf{y}}, \delta_{\max}\}$ guarantees that

$$V^{\text{exact}}(\mathbf{x}_T, \mathbf{y}_T) \leq V_T \leq \varepsilon.$$

Comparison with AGDA. We compare the iteration complexity of TALDA against the $\mathcal{O}(\kappa_{\mathbf{x}}\kappa_{\mathbf{y}}^2 \log(\varepsilon^{-1}))$ complexity of AGDA. The relative efficiency of these algorithms is heavily dictated by the asymmetry of the condition number $(\kappa_{\mathbf{x}}, \kappa_{\mathbf{y}})$ and the strength of the cross-coupling $(\kappa_{\mathbf{x}\mathbf{y}})$. Specifically, TALDA achieves a strictly better complexity when the cross-coupling is relatively weak, yielding an advantage when $\kappa_{\mathbf{x}\mathbf{y}} < \sqrt{\kappa_f}$. AGDA appears more efficient in regimes where the primal variable dominates $(\kappa_{\mathbf{x}} \gg \kappa_{\mathbf{y}})$. However, TALDA's dependence on κ_f is due to the analytical simplifications we made, such as $L = \max\{L_{\mathbf{x}}, L_{\mathbf{y}}\}$ and $\mu = \min\{\mu_{\mathbf{x}}, \mu_{\mathbf{y}}\}$. We conjecture that a more careful analysis that separately deals with these parameters could potentially yield a tighter TALDA bound explicitly in terms of $\kappa_{\mathbf{x}}$ and $\kappa_{\mathbf{y}}$ rather than their maximum. While AGDA

possesses a strictly better logarithmic dependence, this difference in precision scaling ($\log(\varepsilon^{-1})$ vs $\log^2(\varepsilon^{-1})$) is often negligible in practical machine learning applications.

5. Special Cases

We consider several special regimes in which either the Lyapunov function or the underlying min-max problem exhibits additional structure. These settings help clarify how our approach behaves under stronger regularity assumptions. Specifically, we examine (i) the case of a smooth Lyapunov function, which leads to a refined algorithmic variant, (ii) uncoupled min-max problems in which the interaction between players vanishes, recovering standard gradient descent-ascent on the individual objectives, and (iii) an idealized setting with access to exact min-max oracles, where the method achieves an improved convergence rate compared to gradient descent on the idealized Lyapunov.

Smooth Lyapunov Functions. If f satisfies A. 3 and V is L_V -smooth, the update in Algorithm 1 can be refined to explicitly account for second-order coupling by performing a gradient descent step on V :

$$\begin{aligned} \mathbf{x}^{\text{grad}} &= \mathbf{x} - \eta(\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) + \frac{1}{\mu_f} \nabla_{\mathbf{x}\mathbf{y}}^2 f(\mathbf{x}, \hat{\mathbf{y}}) \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}})), \\ \mathbf{y}^{\text{grad}} &= \mathbf{y} + \eta(\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) - \frac{1}{\mu_f} \nabla_{\mathbf{y}\mathbf{x}}^2 f(\hat{\mathbf{x}}, \mathbf{y}) \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y})). \end{aligned}$$

Writing $\kappa_V = L_V/\mu_f$, and after appropriate adaptation of the algorithm parameters, the resulting method (referred to as SOTALDA, see Algorithm 2 in Section D) achieves a complexity bound of $O(\kappa_f \kappa_V \log(\varepsilon^{-1})^2)$.

The resulting dynamics resemble lookahead-type methods such as LOLA (Foerster et al., 2017) and its first-order variants, that incorporate Hessian-vector products to anticipate the effect of one player's update on the other. However, in contrast to LOLA, the correction terms here arise naturally from the Lyapunov descent analysis and are explicitly designed to guarantee a decrease of the Lyapunov function. This highlights how the proposed framework provides a principled foundation for algorithms that leverage second-order interaction terms while retaining convergence guarantees.

Exact min-max oracle. When exact min/max oracles are available, the $\hat{\xi}$ updates in Algorithm 1 reduce to

$$\hat{\mathbf{x}}_{k+1} \in \arg \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}_k).$$

In this case, Equation (15) is satisfied after a single oracle call, with $\delta = 0$. This means the Lyapunov decrease ratio by $1 - r$ triggers at each iteration. Consequently, from Equation (22), the Lyapunov function contracts at the rate

$$V_k \leq \left(1 - \frac{2}{1 + \kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2}\right)^k V_0 = \left(\frac{\kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2 - 1}{\kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2 + 1}\right)^k V_0.$$

This rate is strictly faster than that of gradient descent applied directly to the idealized Lyapunov Equation (8), highlighting that the proposed Lyapunov-based scheme not only matches but can outperform idealized methods, which confirms the tightness of our analysis.

Uncoupled min-max problem. When the objective is uncoupled, i.e., $f(\mathbf{x}, \mathbf{y}) = f_1(\mathbf{x}) - f_2(\mathbf{y})$, we have $L_{\mathbf{x}\mathbf{y}} = 0$. In this setting, the condition in Lemma 3 is trivially satisfied for any choice of $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$. However, the result of Lemma 4 cannot be applied directly as $\kappa_{\mathbf{x}\mathbf{y}} = 0$. To avoid entering the regime where $\Delta_k \geq 0$, we instead enforce $\Delta_k < 0$ by construction. Setting $\delta = 0$ in Equation (17) yields the sufficient condition

$$r V^{\text{Exact}}(\xi_k) \leq \frac{1}{2L_f} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2.$$

Since $\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) = \nabla f_1(\mathbf{x})$ and $\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) = -\nabla f_2(\mathbf{y})$, choosing $r = \mu_f/L_f$ ensures that $\Delta_k < 0$ uniformly, and we recover the standard linear convergence rate of gradient descent on smooth PL functions. This behavior is expected: in the uncoupled case, Algorithm 1 reduces to gradient descent on $\mathbf{x}$ and gradient ascent on $\mathbf{y}$ with step size $1/L_f$.

6. Experimental Results

In our experiments, we focus on comparisons with alternating gradient descent-ascent (AGDA). To the best of our knowledge, AGDA is the only first-order method for which linear convergence guarantees are available under S2PL, making it the most relevant baseline in this setting.

6.1. Practical guidelines for setting TALDA hyperparameters

In many practical settings, the exact constants that TALDA uses (L_f , $\kappa_{\mathbf{x}\mathbf{y}}$, $L_{\mathbf{x}\mathbf{y}}$) are rarely known a priori. However, they can be chosen adaptively while preserving the theoretical convergence rates up to constant factors. For instance, instead of fixing the step size as $\hat{\eta} = \frac{1}{L_f}$ in lines 6-7 of Algorithm 1, one may select $\hat{\eta}_k$ such that

$$\begin{aligned} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k) &= f(\hat{\mathbf{x}}_k - \hat{\eta}_k \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}_k, \mathbf{y}_k), \mathbf{y}_k) \\ &\leq f(\hat{\mathbf{x}}_k, \mathbf{y}_k) - \frac{\hat{\eta}_k}{2} \|\nabla_{\mathbf{x}} f(\hat{\mathbf{x}}_k, \mathbf{y}_k)\|^2, \end{aligned}$$

and similarly for $\hat{\mathbf{y}}_k$. This condition corresponds to a simple backtracking line search, and it preserves the same theoretical convergence guarantees as the fixed-step variant up to a factor of 2. A similar backtracking procedure can also be used to estimate $L_{\mathbf{x}\mathbf{y}}$. In addition, the primary condition for the convergence of TALDA is maintaining $\eta < \hat{\eta}$ (typically, $\eta = O((1 - \delta)\hat{\eta})$), that can be found by grid search if $\eta = (1 - \delta) \max_k \{\hat{\eta}_k\}$ does not work.

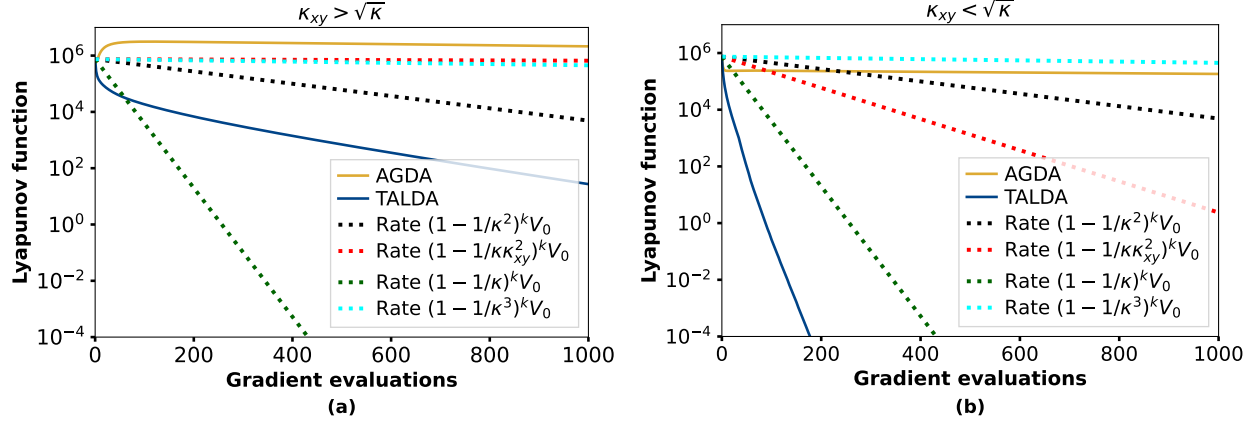


Figure 1. Comparing the performance of TALDA and AGDA in two different regimes of condition numbers κ and κ_{xy} . We construct the quadratic problem with size $d_x = 50$, $d_y = 100$. For (a), we set $\kappa = 10$ and $\kappa_{xy} = 20$. For (b), we set $\kappa = 10$ and $\kappa_{xy} = 2$. The parameters for TALDA and AGDA are chosen according to their theoretically prescribed values. Red, green, and black dotted lines denote different theoretical rates.

This naturally aligns with our theoretical derivation, where $\eta = (1 - \delta)/(L_f + L_{xy}\kappa_{xy})$, which is strictly less than $\frac{1}{L_f}$. Furthermore, because κ_{xy} depends on the strong convexity parameter μ_f , κ_{xy} can typically be estimated directly from the user-specified regularization parameter of the problem.

The parameter δ is user-defined, and empirical results show that setting it to a small positive value (e.g., 0.1) yields strong convergence. Finally, the parameter r , which governs the rate of decrease for the Lyapunov function, can be set to any real number between 0 and 1 (preferably smaller, say $r = 1e^{-3}$), and the algorithm will converge.

6.2. Quadratic function

To isolate and illustrate the qualitative differences in convergence behavior predicted by our theory, we evaluate TALDA and AGDA on the following synthetic minimax problem

$$\min_{\mathbf{x}} \max_{\mathbf{y}} \frac{1}{2} \mathbf{x}^T A \mathbf{x} + \mathbf{y}^T B \mathbf{x} - \frac{1}{2} \mathbf{y}^T C \mathbf{y} + d^T \mathbf{x} - e^T \mathbf{y}. \quad (23)$$

We generated matrices $A \in \mathbb{R}^{d_x \times d_x}$ and $C \in \mathbb{R}^{d_y \times d_y}$ from Gaussian distributions $\mathcal{N}(0, \Sigma_{d_x})$ and $\mathcal{N}(0, \Sigma_{d_y})$, respectively, and adjusted their spectra so that the eigenvalues were linearly spaced within $[\mu_x, L_x]$ and $[\mu_y, L_y]$. This ensures the resulting functions satisfy S2PL, and admit a saddle point and thus provides an appropriate testbed for assessing whether the theoretical convergence rates of the proposed algorithm are reflected in practice.

In Figure 1, we report the performance of both methods under two regimes: $\kappa_{xy} > \sqrt{\kappa_f}$ and $\kappa_{xy} < \sqrt{\kappa_f}$. Learning rates for AGDA and TALDA are chosen according to theoretical prescriptions from Yang et al. (2020) and the step size η in Equation (16), respectively. Consistent with our theory, when $\kappa_{xy} < \sqrt{\kappa_f}$, TALDA significantly outperforms

AGDA, as shown in Figure 1(b). Conversely, in the strongly coupled regime $\kappa_{xy} > \sqrt{\kappa_f}$, TALDA and AGDA share the same theoretical guarantees, but empirically TALDA outperform AGDA, as illustrated in Figure 1(a). In both cases, TALDA often exceeds the predicted theoretical performance, which can be attributed to its adaptive design.

6.3. Robust Least Squares

The robust least squares (RLS) problem minimizes the worst-case squared error under bounded perturbations in the observed data and is given by $\min_{\mathbf{x}} \max_{\delta: \|\delta\| \leq \rho} \|A\mathbf{x} - \mathbf{y}\|^2$ where $A \in \mathbb{R}^{n \times m}$ is a coefficient matrix, $\mathbf{y}_0 \in \mathbb{R}^n$ is the observed noisy vector with bounded deterministic perturbation $\delta = \mathbf{y}_0 - \mathbf{y}$. A relaxed version of RLS, referred to as RLS with a soft constraint, is given by

$$\min_{\mathbf{x}} \max_{\mathbf{y}} \|A\mathbf{x} - \mathbf{y}\|_M^2 - \lambda \|\mathbf{y} - \mathbf{y}_0\|_M^2, \quad (24)$$

where $0 \leq M$ and $\|\mathbf{x}\|_M^2 = \mathbf{x}^T M \mathbf{x}$. Unlike standard ridge regression, which penalizes the model parameters $\mathbf{x}$ to regularize complexity, RLS with a soft constraint applies an L2 penalty strictly to the target variable $\mathbf{y}$ to model bounded uncertainty in the adversary's action space. It can be decomposed into the sum of a strongly convex-strongly concave function and an affine term, and therefore satisfies the two-sided PL condition with $\lambda > 1$. Leveraging this property, we evaluate the performance of our proposed methods against AGDA on this problem.

Dataset Generation We generate the dataset for the RLS problem similarly to Du & Hu (2019); Yang et al. (2020). For the sake of completeness, we briefly describe the data generation process. We set $M = I_n$, $\lambda = 3$, $n = 1000$, $m = 500$, $\mathbf{y}_0 = A\mathbf{x}^* + \epsilon$ and sample $\mathbf{x}^*$ from $\mathcal{N}(0, I_m)$, the rows of A from $\mathcal{N}(0, I_n)$ and ϵ from $\mathcal{N}(0, 0.01)$.

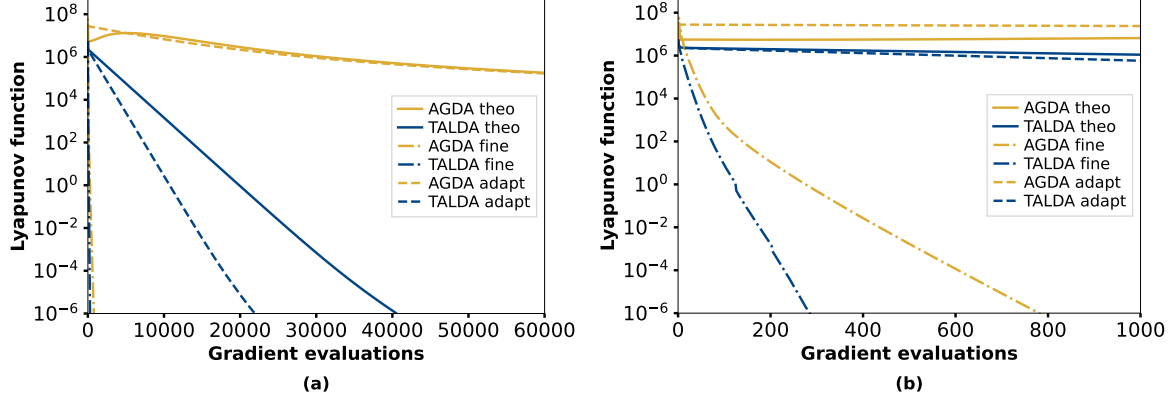


Figure 2. TALDA and AGDA convergence on Robust Least Squares ($L_x = 5697.3$, $L_y = 4.0$, $L_{xy} = 106.7$, $\mu_x = 169.5$, $\mu_y = 4.0$) across theoretical, adaptive, and fine-tuned hyperparameter settings. (a) Full 60,000 evaluation horizon. (b) Zoomed-in view of (a) for the first 1,000 evaluations.

Observation Figure 2 compares AGDA and TALDA across three learning rate regimes: theoretically prescribed, adaptive (dynamically estimating L_f , L_{xy} , etc., as in Section 6.1), and empirically fine-tuned. For fine-tuning, we conduct a grid search to find the best-performing combination of learning rates for the x and y updates of both algorithms. Across all settings, TALDA consistently outperforms AGDA, demonstrating robustness beyond the theoretical regime. With theoretical step sizes, AGDA initially exhibits apparent divergence in the Lyapunov values, reflecting the non-monotonic behavior often encountered in function-value-based analyses. Convergence guarantees of the form $f(\mathbf{x}_k) - f^* \leq C(1 - \rho)^k$ allow for substantial transient growth, which is absorbed into the constant C . Here, large κ_x leads to a slow rate and a prolonged transient phase, requiring many iterations for AGDA to recover from this local divergence.

7. Conclusion and Future Work

In this work, we present a principled framework for designing algorithms for smooth nonconvex-nonconcave minimax problems that satisfy the two-sided Polyak-Lojasiewicz (PL) condition. By using Lyapunov functions as design principles rather than analysis tools, we developed a methodology that explicitly builds stability and convergence into algorithms. Using this methodology, we propose a new single-loop algorithm, TALDA, that alternates between refining the Lyapunov approximation and performing gradient updates. We established that TALDA achieves linear convergence with an iteration complexity of $O((\kappa_f^2 + \kappa_f)\kappa_{xy}^2 \log^2(1/\epsilon))$. Crucially, our analysis explicitly captures the interaction between players through the cross-smoothness constant κ_{xy} . This yields better convergence rates in weakly coupled regimes ($\kappa_{xy} < \sqrt{\kappa_f}$) compared to standard approaches like AGDA, which typically scale with κ_y^2 regardless of coupling strength. Experimental results on synthetic quadratic

games and robust least squares problems confirm our theoretical insights. TALDA significantly outperforms AGDA in weakly coupled scenarios while matching the best-known asymptotic rates in strongly coupled regimes, demonstrating the robustness of Lyapunov-guided design in diverse problem landscapes.

The Lyapunov-guided design methodology provides a strong foundation for several promising extensions. Firstly, this framework can be extended to stochastic settings by replacing Lyapunov decrease with expected Lyapunov decrease and drawing inspiration from stabilizing control via Stochastic Lyapunov techniques (Deng & Krstić, 1997; Florchinger, 1995). Secondly, the methodology can be extended to settings beyond two-sided PL by identifying a computable gap or Lyapunov function in that setting. For instance, in constrained convex-concave settings, one could utilize the Frank-Wolfe linearization gap as a computable candidate (Gidel et al., 2017, Equation 7). Similarly, generic gap functions developed for monotone variational inequalities in Zhu & Marcotte (1994); Wu et al. (1993) can serve as inspiration for Lyapunov candidates in those domains. Finally, applying Catalyst-style acceleration techniques (Chen et al., 2022) to the TALDA updates offers an exciting pathway for achieving accelerated convergence rates while maintaining our improved dependence on the cross-smoothness parameter.

By bridging dynamical systems theory and algorithm design, this work serves as a foundational step for a principled Lyapunov-based framework for designing min-max algorithms. In smooth two-sided PL setting, this methodology produced a non-standard algorithm like TALDA, where unique mechanisms such as the variable swap arise directly alongside guaranteed proofs of convergence. Extending this design framework to stochastic and other relaxed settings offers an exciting frontier that promises to uncover new algorithmic structures and mathematical techniques.

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Impact Statement

This work advances the theory of nonconvex-nonconcave min-max optimization, with a primary focus on providing a principled framework for algorithm design. As a primarily theoretical contribution, its societal impact is indirect, and more work is needed for this work to be used at scale in downstream applications. Here, we highlight the potential positive and negative impacts of this work in the field of machine learning and AI.

The proposed algorithm, TALDA, and the associated theoretical framework directly address the instability often observed in training minimax problems. A key application of this work is adversarial machine learning and robust optimization. By providing algorithms with linear convergence guarantees in these settings, this research contributes to the development of AI systems that are more resistant to adversarial attacks and distributional shifts. This is critical for the deployment of machine learning in safety-critical domains where reliability is paramount. This improvement in iteration complexity can lead to reduced computational resources and energy consumption required to train complex models, aligning with the goals of sustainable computing.

Since the training of Generative Adversarial Networks (GANs) is modeled as a minimax game, this work can potentially be used to develop better algorithms for generative AI. While improved stability in GAN training accelerates creative and scientific applications (e.g., drug discovery, synthesis), it also lowers the barrier to creating high-fidelity synthetic media. This could potentially exacerbate societal challenges related to deepfakes, disinformation campaigns, and the erosion of trust in digital media. We encourage the community to continue developing detection methods and technologies alongside generation capabilities to mitigate these risks.

Overall, this work aims to transform algorithm design in minimax optimization into a principled approach. We believe the benefits of enabling more robust, verifiable, and energy-efficient AI systems outweigh the risks, provided that the capabilities of generative models are monitored responsibly.

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A. Related work

A.1. Strongly-convex-strongly-concave setting

A growing body of work has studied min-max optimization in the smooth, strongly convex-strongly concave (SCSC) setting. Classical approaches include gradient-based methods, such as extra-gradient and simultaneous gradient descent-ascent, which achieve linear convergence under strong convexity-strong concavity assumptions. Extragradient (Korpelevich, 1976; Gidel et al., 2019a; Tseng, 1995; Mokhtari et al., 2020) in this setting achieves $\tilde{O}(\kappa_x + \kappa_y)$ in the number of gradient evaluations. Simultaneous gradient descent-ascent (Lin et al., 2020a) achieves $\tilde{O}(\kappa_x \kappa_y)$ whereas alternating gradient-descent-ascent (Lee et al., 2024; Zhang et al., 2022a) achieves $\tilde{O}(\kappa_x + \kappa_y + \kappa_{xy}(\sqrt{\kappa_x} + \sqrt{\kappa_y}))$ number of gradient evaluations. Recently, the lower complexity bound was studied in this setting and was found to be $\tilde{\Omega}(\sqrt{\kappa_x \kappa_y})$ (Ibrahim et al., 2020; Zhang et al., 2022b). This lower bound was achieved by Lin et al. (2020b) and Kovalev & Gasnikov (2022). Although this work focuses primarily on convergence, we believe that our approach may yield algorithms with optimal rates.

A.2. Lyapunov-related

Lyapunov functions are mostly used as a proof technique to show the convergence of a given algorithm. We will discuss important recent works that use Lyapunov functions for convergence and stability analysis in min-max games. Zheng et al. (2023) use a novel Lyapunov function to show convergence of their proposed algorithm for nonconvex nonconcave min-max optimization problems. They use the descent property of the Lyapunov function to inform the choice of their hyperparameters. Vlatakis-Gkaragkounis et al. (2021) construct an initialization-dependent Lyapunov function to provide convergence and stability analysis of the corresponding algorithm for solving a class of optimization problems called hidden convex-concave games (HCC games). Hemmat et al. (2023) propose a new optimizer for min-max games called LEAD and study its convergence properties for quadratic games using Lyapunov stability theory.

Recently, some works have started exploring Lyapunov functions as a design tool in optimization. Du (2022) propose a two-phase systematic framework using Lyapunov functions to design new approximation algorithms. The first phase focuses on designing continuous-time approximation algorithms with provable approximation ratios, while the second phase converts these algorithms to discrete-time algorithms with almost the same approximation ratio and provable time complexity. There are two major differences between Du (2022) and our work. In our study, i) we are designing stable algorithms for min-max optimization problems; ii) secondly, we engage directly with discrete time, thereby circumventing the challenge of preserving continuous time convergence guarantees when transitioning to discrete time.

Upadhyaya et al. (2025) use Lyapunov functions to design discrete-time algorithms for monotone problems. Similar to the computable Lyapunov function we used in Equation (10), Diakonikolas & Wang (2022) uses a similar family of Lyapunov functions to unify the analysis of algorithms in convex and minmax optimization. The framework provides an intuitive interpretation of the mechanism driving the convergence as a trade-off between reducing the norm of the gradient and reducing some notion of optimality gap. Gramlich et al. (2020) convert the Lyapunov functions based algorithm design problem into a semi-definite program and show how this convex formulation enables the systematic synthesis of accelerated optimization methods for minimization and saddle point problems.

Definition 3. A function $f : \mathbb{R}^{d_x} \times \mathbb{R}^{d_y} \rightarrow \mathbb{R}$ is Lipschitzian smooth if $\exists L_{xx}, L_{xy}, L_{yx}, L_{yy} > 0$ such that $\forall \mathbf{x}, \mathbf{x}' \in \mathbb{R}^{d_x}, \forall \mathbf{y}, \mathbf{y}' \in \mathbb{R}^{d_y}$ the following holds true:

$$\begin{aligned} \|\nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f(\mathbf{x}', \mathbf{y})\| &\leq L_{xx} \|\mathbf{x} - \mathbf{x}'\| & \|\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}')\| &\leq L_{yy} \|\mathbf{y} - \mathbf{y}'\| \\ \|\nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}')\| &\leq L_{xy} \|\mathbf{y} - \mathbf{y}'\| & \|\nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}) - \nabla_{\mathbf{y}} f(\mathbf{x}', \mathbf{y})\| &\leq L_{yx} \|\mathbf{x} - \mathbf{x}'\|. \end{aligned}$$

This implies that the following inequalities hold true:

$$\frac{-L_{xx}}{2} \|\mathbf{x}' - \mathbf{x}\|^2 \leq f(\mathbf{x}', \mathbf{y}) - f(\mathbf{x}, \mathbf{y}) - \langle \nabla_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}), \mathbf{x}' - \mathbf{x} \rangle \leq \frac{L_{xx}}{2} \|\mathbf{x}' - \mathbf{x}\|^2 \quad (25)$$

$$\frac{-L_{yy}}{2} \|\mathbf{y}' - \mathbf{y}\|^2 \leq f(\mathbf{x}, \mathbf{y}') - f(\mathbf{x}, \mathbf{y}) - \langle \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y}), \mathbf{y}' - \mathbf{y} \rangle \leq \frac{L_{yy}}{2} \|\mathbf{y}' - \mathbf{y}\|^2. \quad (26)$$

B. Technical Lemmas

This section contains Lemmas required for the analysis that do not appear in the main paper.

Lemma 5. *Let V be defined as in Equation (7). Under A. 1 and 2, V is a Lyapunov candidate function in the sense of Def. 1.*

Proof. Recall $V(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}, \mathcal{Y}(\mathbf{x})) - f(\mathcal{X}(\mathbf{y}), \mathbf{y})$, where $\mathcal{X}(\mathbf{y}) \in \arg \min_{\mathbf{x}'} f(\mathbf{x}', \mathbf{y})$ and $\mathcal{Y}(\mathbf{x}) \in \arg \max_{\mathbf{y}'} f(\mathbf{x}, \mathbf{y}')$.

Nonnegativity and characterization of the zero set. By optimality of $\mathcal{X}(\mathbf{y})$ and $\mathcal{Y}(\mathbf{x})$,

$$f(\mathbf{x}, \mathcal{X}(\mathbf{y})) \geq f(\mathbf{x}, \mathbf{y}) \quad \text{and} \quad f(\mathcal{Y}(\mathbf{x}), \mathbf{y}) \leq f(\mathbf{x}, \mathbf{y}).$$

Subtracting yields $V(\mathbf{x}, \mathbf{y}) \geq 0$. Moreover, $V(\mathbf{x}, \mathbf{y}) = 0$ implies

$$f(\mathbf{x}, \mathcal{X}(\mathbf{y})) = f(\mathbf{x}, \mathbf{y}) = f(\mathcal{Y}(\mathbf{x}), \mathbf{y}),$$

so $\mathbf{y} \in \arg \max_{\mathbf{y}'} f(\mathbf{x}, \mathbf{y}')$ and $\mathbf{x} \in \arg \min_{\mathbf{x}'} f(\mathbf{x}', \mathbf{y})$; i.e., $(\mathbf{x}, \mathbf{y})$ is a saddle point. Conversely, if $(\mathbf{x}, \mathbf{y})$ is a saddle point, then $\mathbf{y} \in \arg \max_{\mathbf{y}'} f(\mathbf{x}, \mathbf{y}')$ and $\mathbf{x} \in \arg \min_{\mathbf{x}'} f(\mathbf{x}', \mathbf{y})$, so $V(\mathbf{x}, \mathbf{y}) = 0$.

Strict positivity off the equilibrium set. By Lemma 1, under A. 1 the saddle points coincide with the stationary points of f (equivalently, with the global minimax points). Therefore, if $(\mathbf{x}, \mathbf{y}) \notin \mathcal{S}$ (the set of saddle points), then it is not a stationary point, and in particular it cannot satisfy simultaneously $\mathbf{x} \in \mathcal{X}(\mathbf{y})$ and $\mathbf{y} \in \mathcal{Y}(\mathbf{x})$. Hence at least one of the inequalities $f(\mathbf{x}, \hat{\mathbf{y}}) \geq f(\mathbf{x}, \mathbf{y})$ or $f(\mathbf{x}, \mathbf{y}) \geq f(\hat{\mathbf{x}}, \mathbf{y})$ is strict, which implies $V(\mathbf{x}, \mathbf{y}) > 0$.

Thus, $V(\xi) > 0$ for all $\xi \notin \mathcal{S}$ and $V(\xi) = 0$ iff $\xi \in \mathcal{S}$, so V is a Lyapunov candidate function as in Def. 1. $\square$

Lemma 6. *Let V be defined as in Equation (10). Under A. 1 and 2, V is a Lyapunov candidate function (Def. 1).*

Proof. By Lemma 2, for all $(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}})$,

$$V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) = f(\mathcal{X}(\mathbf{y}), \mathbf{y}) - f(\mathbf{x}, \mathcal{Y}(\mathbf{x})) \leq V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}).$$

Since V^{Exact} is a Lyapunov candidate function by Lemma 5, it follows immediately that $V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \geq 0$ for all arguments.

Moreover, if $(\mathbf{x}, \mathbf{y})$ is a saddle point, then by definition $\mathbf{x} = \mathcal{X}(\mathbf{y})$ and $\mathbf{y} = \mathcal{Y}(\mathbf{x})$, and therefore $V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) = 0$. Choosing $\tilde{\mathbf{x}} = \mathcal{X}(\mathbf{y})$ and $\tilde{\mathbf{y}} = \mathcal{Y}(\mathbf{x})$ further yields $\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}, \mathbf{y}) = 0$ and $\nabla_{\mathbf{y}} f(\mathbf{x}, \tilde{\mathbf{y}}) = 0$ under A. 1, so that $V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = 0$.

Conversely, if $V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = 0$, then necessarily $V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) = 0$, which implies $(\mathbf{x}, \mathbf{y})$ is a saddle point by Lemma 1. Since V is a sum of nonnegative terms, equality also forces $\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}, \mathbf{y}) = 0$ and $\nabla_{\mathbf{y}} f(\mathbf{x}, \tilde{\mathbf{y}}) = 0$, hence $\tilde{\mathbf{x}} = \mathcal{X}(\mathbf{y})$ and $\tilde{\mathbf{y}} = \mathcal{Y}(\mathbf{x})$.

Therefore, V is nonnegative and vanishes if and only if the state corresponds to a saddle point with consistent best responses, which proves that V is a Lyapunov candidate function. $\square$

Lemma 7. *Let g be a twice differentiable function whose Hessian is Lipschitz continuous, i.e.,*

$$\|\nabla^2 g(\mathbf{y}) - \nabla^2 g(\mathbf{x})\| \leq L_g \|\mathbf{y} - \mathbf{x}\|.$$

Then, if $h = 2\epsilon/L_g \|\mathbf{v}\|^2$, we have

$$\left\| \nabla^2 g(\mathbf{x}) \mathbf{v} - \left(\frac{\nabla g(\mathbf{x} + h\mathbf{v}) - \nabla g(\mathbf{x})}{h} \right) \right\| \leq \epsilon.$$

Proof. Define $\phi(t) := \nabla g(\mathbf{x} + t\mathbf{v})$ for $t \in [0, h]$. By the fundamental theorem of calculus,

$$\frac{\nabla g(\mathbf{x} + h\mathbf{v}) - \nabla g(\mathbf{x})}{h} = \frac{1}{h} \int_0^h \phi'(t) dt = \frac{1}{h} \int_0^h \nabla^2 g(\mathbf{x} + t\mathbf{v}) \mathbf{v} dt.$$

Hence

$$\frac{\nabla g(\mathbf{x} + h\mathbf{v}) - \nabla g(\mathbf{x})}{h} - \nabla^2 g(\mathbf{x}) \mathbf{v} = \frac{1}{h} \int_0^h (\nabla^2 g(\mathbf{x} + t\mathbf{v}) - \nabla^2 g(\mathbf{x})) \mathbf{v} dt.$$

Taking norms and using submultiplicativity,

$$\left\| \frac{\nabla g(\mathbf{x} + h\mathbf{v}) - \nabla g(\mathbf{x})}{h} - \nabla^2 g(\mathbf{x})\mathbf{v} \right\| \leq \frac{1}{h} \int_0^h \|\nabla^2 g(\mathbf{x} + t\mathbf{v}) - \nabla^2 g(\mathbf{x})\| \|\mathbf{v}\| dt.$$

By L_g -Lipschitzness of the Hessian,

$$\|\nabla^2 g(\mathbf{x} + t\mathbf{v}) - \nabla^2 g(\mathbf{x})\| \leq L_g \|t\mathbf{v}\| = L_g t \|\mathbf{v}\|.$$

Therefore,

$$\left\| \frac{\nabla g(\mathbf{x} + h\mathbf{v}) - \nabla g(\mathbf{x})}{h} - \nabla^2 g(\mathbf{x})\mathbf{v} \right\| \leq \frac{1}{h} \int_0^h L_g t \|\mathbf{v}\|^2 dt = \frac{L_g}{h} \frac{h^2}{2} \|\mathbf{v}\|^2 = \frac{L_g}{2} h \|\mathbf{v}\|^2.$$

Choosing $h = \frac{2\epsilon}{L_g \|\mathbf{v}\|^2}$ gives the desired bound $\leq \epsilon$. $\square$

Theorem 2 (Linear convergence of gradient descent on smooth, PL functions (Karimi et al., 2016)). *Let $\phi : \mathbb{R}^d \rightarrow \mathbb{R}$ be differentiable and satisfy the smoothness and PL conditions in (2) with constants L and μ . Define gradient descent with step size $1/L$ by $\mathbf{x}_{k+1} = \mathbf{x}_k - \frac{1}{L} \nabla \phi(\mathbf{x}_k)$. Then, for all $k \geq 0$,*

$$\phi(\mathbf{x}_k) - \phi^* \leq \left(1 - \frac{\mu}{L}\right)^k (\phi(\mathbf{x}_0) - \phi^*), \quad \text{where } \phi^* := \min_{\mathbf{x}} \phi(\mathbf{x}).$$

Proof. Let $\phi^* := \min_{\mathbf{x}} \phi(\mathbf{x})$. By L -smoothness (first inequality in (2)), for any $\mathbf{x}$ and any update $\mathbf{x}^+$,

$$\phi(\mathbf{x}^+) \leq \phi(\mathbf{x}) + \langle \nabla \phi(\mathbf{x}), \mathbf{x}^+ - \mathbf{x} \rangle + \frac{L}{2} \|\mathbf{x}^+ - \mathbf{x}\|^2.$$

Apply this with $\mathbf{x}_k$ and $\mathbf{x}^+ = \mathbf{x}_k - \frac{1}{L} \nabla \phi(\mathbf{x}_k)$ yields

$$\phi(\mathbf{x}_{k+1}) \leq \phi(\mathbf{x}_k) - \frac{1}{2L} \|\nabla \phi(\mathbf{x}_k)\|^2.$$

Using the PL inequality (second inequality in (2)), $\|\nabla \phi(\mathbf{x}_k)\|^2 \geq 2\mu(\phi(\mathbf{x}_k) - \phi^*)$, hence

$$\phi(\mathbf{x}_{k+1}) - \phi^* \leq \left(1 - \frac{\mu}{L}\right) (\phi(\mathbf{x}_k) - \phi^*).$$

Iterating the recursion yields the stated bound. $\square$

Corollary 1 (Iteration complexity for objective gap and gradient norm). *Under the assumptions of Theorem 2:*

$$\begin{aligned} \phi(\mathbf{x}_k) - \phi^* < \epsilon &\iff k \geq \frac{L}{\mu} \log\left(\frac{\phi(\mathbf{x}_0) - \phi^*}{\epsilon}\right), \\ \|\nabla \phi(\mathbf{x}_k)\|^2 < \epsilon_{\nabla} &\iff k \geq \frac{L}{\mu} \log\left(\frac{2L(\phi(\mathbf{x}_0) - \phi^*)}{\epsilon_{\nabla}}\right) \iff k \geq \frac{L}{\mu} \log\left(\frac{L \|\nabla f(x_0)\|^2}{\mu \epsilon_{\nabla}}\right). \end{aligned}$$

Proof. The first claim is immediate from Theorem 2 by requiring $(1 - \frac{\mu}{L})^k (\phi(\mathbf{x}_0) - \phi^*) \leq \epsilon$ and solving for k .

For the second claim, use the descent inequality from the proof of Theorem 2:

$$\phi(\mathbf{x}_{k+1}) \leq \phi(\mathbf{x}_k) - \frac{1}{2L} \|\nabla \phi(\mathbf{x}_k)\|^2,$$

which implies $\|\nabla \phi(\mathbf{x}_k)\|^2 \leq 2L(\phi(\mathbf{x}_k) - \phi(\mathbf{x}_{k+1})) \leq 2L(\phi(\mathbf{x}_k) - \phi^*)$. Combining with Theorem 2 yields

$$\|\nabla \phi(\mathbf{x}_k)\|^2 \leq 2L \left(1 - \frac{\mu}{L}\right)^k (\phi(\mathbf{x}_0) - \phi^*),$$

and the result follows by solving $2L \left(1 - \frac{\mu}{L}\right)^k (\phi(\mathbf{x}_0) - \phi^*) \leq \epsilon_{\nabla}$ for k .

In both claims, we get the simplified result by using the bound $-\log(1 - t) \geq t$ for $t \in (0, 1)$ as follow,

$$\log\left(\frac{1}{1 - \mu/L}\right) \geq \frac{L}{\mu}.$$

$\square$

C. Missing Proofs

In this section, we provide the proofs for all Lemmas and Theorems stated in the main text. We restate each result before its proof for readability.

C.1. Proof of Lemma 2

Lemma 2. *If f satisfies A. 1 and 2, the Lyapunov functions in Equations (7) and (10) satisfy, $\forall (\mathbf{x}, \mathbf{y}) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$,*

$$V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) \leq V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \quad \forall (\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}. \quad (11)$$

In addition, $V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) = \min_{\tilde{\mathbf{x}}, \tilde{\mathbf{y}}} V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}})$.

Proof. Fix $(\mathbf{x}, \mathbf{y})$ and $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$. By definition,

$$V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = f(\mathbf{x}, \tilde{\mathbf{y}}) - f(\tilde{\mathbf{x}}, \mathbf{y}) + \frac{1}{2\mu} \left(\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}, \mathbf{y})\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}, \tilde{\mathbf{y}})\|^2 \right).$$

Under A. 1, applying the PL inequality (2) to $f(\cdot, \mathbf{y})$ at $\tilde{\mathbf{x}}$ yields

$$f(\tilde{\mathbf{x}}, \mathbf{y}) - f(\mathcal{X}(\mathbf{y}), \mathbf{y}) \leq \frac{1}{2\mu} \|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}, \mathbf{y})\|^2,$$

and applying it to $-f(\mathbf{x}, \cdot)$ at $\tilde{\mathbf{y}}$ yields

$$f(\mathbf{x}, \mathcal{Y}(\mathbf{x})) - f(\mathbf{x}, \tilde{\mathbf{y}}) \leq \frac{1}{2\mu} \|\nabla_{\mathbf{y}} f(\mathbf{x}, \tilde{\mathbf{y}})\|^2.$$

Adding these two inequalities gives

$$f(\mathbf{x}, \tilde{\mathbf{y}}) - f(\tilde{\mathbf{x}}, \mathbf{y}) + \frac{1}{2\mu} \left(\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}, \mathbf{y})\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}, \tilde{\mathbf{y}})\|^2 \right) \geq f(\mathbf{x}, \mathcal{Y}(\mathbf{x})) - f(\mathcal{X}(\mathbf{y}), \mathbf{y}) = V^{\text{Exact}}(\mathbf{x}, \mathbf{y}),$$

which proves $V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) \leq V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}})$ for all $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$.

For the second claim, choose $\tilde{\mathbf{x}} = \mathcal{X}(\mathbf{y})$ and $\tilde{\mathbf{y}} = \mathcal{Y}(\mathbf{x})$. Then the two PL gaps vanish and the gradient terms are zero, so $V(\mathbf{x}, \mathbf{y}, \mathcal{X}(\mathbf{y}), \mathcal{Y}(\mathbf{x})) = V^{\text{Exact}}(\mathbf{x}, \mathbf{y})$. Together with the global lower bound above, this implies $V^{\text{Exact}}(\mathbf{x}, \mathbf{y}) = \min_{\tilde{\mathbf{x}}, \tilde{\mathbf{y}}} V(\mathbf{x}, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}})$. $\square$

C.2. Proof of Lemma 3

Lemma 3. *Under A. 1, if $\hat{\xi} = (\hat{\mathbf{x}}, \hat{\mathbf{y}})$ satisfies for $\delta \in [0, 1]$*

$$\left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \\ \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \end{bmatrix} \right\| \leq \frac{\delta}{\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \\ \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \end{bmatrix} \right\|, \quad (15)$$

then the gradient step $\xi^{\text{grad}} = (\mathbf{x}^{\text{grad}}, \mathbf{y}^{\text{grad}})$ on f defined by

$$\begin{aligned} \mathbf{x}^{\text{grad}} &= \mathbf{x} - \eta \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \mathbf{y}^{\text{grad}} &= \mathbf{y} + \eta \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{aligned}, \quad \eta = \frac{1 - \delta}{L_f + L_{\mathbf{xy}} \kappa_{\mathbf{xy}}} \quad (16)$$

guarantees that the Lyapunov decrease satisfies

$$V(\xi, \hat{\xi}) - V(\xi^{\text{grad}}, \hat{\xi}) \geq \frac{(1 - \delta)^2 / 2}{L_f + L_{\mathbf{xy}} \kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2.$$

Proof. Define

$$\begin{bmatrix} \mathbf{x}^\eta \\ \mathbf{y}^\eta \end{bmatrix} = \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} - \eta \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ -\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix}.$$

By smoothness, the descent lemma tells us that, for $\eta > 0$,

$$f(\mathbf{x}^\eta, \hat{\mathbf{y}}) \leq f(\mathbf{x}, \hat{\mathbf{y}}) - \eta \left(1 - \frac{L_f}{2}\eta\right) \|\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})\|^2 \quad (27)$$

$$-f(\hat{\mathbf{x}}, \mathbf{y}^\eta) \leq -f(\hat{\mathbf{x}}, \mathbf{y}) - \eta \left(1 - \frac{L_f}{2}\eta\right) \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y})\|^2. \quad (28)$$

Therefore, adding the two inequalities alongside with $\frac{1}{2\mu} (\|\nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}})\|^2)$ gives us

$$\begin{aligned} V(\mathbf{x}^\eta, \mathbf{y}^\eta, \hat{\mathbf{x}}, \hat{\mathbf{y}}) &\leq f(\mathbf{x}, \hat{\mathbf{y}}) - f(\hat{\mathbf{x}}, \mathbf{y}) - \eta \left(1 - \frac{L_f}{2}\eta\right) (\|\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})\|^2 + \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y})\|^2) \\ &\quad + \frac{1}{2\mu} (\|\nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}})\|^2), \end{aligned}$$

which can be rewritten as

$$\begin{aligned} V(\mathbf{x}^\eta, \mathbf{y}^\eta, \hat{\mathbf{x}}, \hat{\mathbf{y}}) &\leq f(\mathbf{x}, \hat{\mathbf{y}}) - f(\hat{\mathbf{x}}, \mathbf{y}) - \eta \left(1 - \frac{L_f}{2}\eta\right) \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 + \frac{1}{2\mu} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta) \\ \nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}}) \end{bmatrix} \right\|^2 \\ &= V(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}) - \eta \left(1 - \frac{L_f}{2}\eta\right) \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 + \frac{1}{2\mu} \left(\left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta) \\ \nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}}) \end{bmatrix} \right\|^2 - \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \\ \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \end{bmatrix} \right\|^2 \right). \end{aligned}$$

By cross smoothness,

$$\begin{aligned} \|\nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}}) - \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}})\| &\leq L_{\mathbf{xy}} \eta \|\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})\| \\ \|\nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta) - \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y})\| &\leq L_{\mathbf{xy}} \eta \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y})\|, \end{aligned}$$

which implies

$$\begin{aligned} \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta) \end{bmatrix} \right\| &\leq \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\| + \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}}) - \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta) - \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\| \\ &\leq \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\| + L_{\mathbf{xy}} \eta \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|. \end{aligned}$$

Hence,

$$\left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta) \end{bmatrix} \right\|^2 - \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 \leq L_{\mathbf{xy}} \eta \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\| \left(2 \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\| + L_{\mathbf{xy}} \eta \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\| \right).$$

Using Equation (15)

$$\left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}^\eta, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}^\eta) \end{bmatrix} \right\|^2 - \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 \leq L_{\mathbf{xy}} \eta \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 \left(2 \frac{\delta}{\kappa_{\mathbf{xy}}} + L_{\mathbf{xy}} \eta \right).$$

Therefore, the bound on the Lyapunov becomes

$$\begin{aligned} V(\mathbf{x}^\eta, \mathbf{y}^\eta, \hat{\mathbf{x}}, \hat{\mathbf{y}}) &\leq V(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}) - \eta \left(1 - \frac{L_f}{2}\eta\right) \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 + \frac{1}{2\mu} L_{\mathbf{xy}} \eta \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 \left(2 \frac{\delta}{\kappa_{\mathbf{xy}}} + L_{\mathbf{xy}} \eta \right) \\ &= V(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}) + \left(-\eta \left(1 - \frac{L_f}{2}\eta\right) + \eta \left(\delta + \frac{\kappa_{\mathbf{xy}} L_{\mathbf{xy}}}{2} \eta \right) \right) \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2 \\ &= V(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}) + \left(\frac{\eta^2}{2} (L_f + \kappa_{\mathbf{xy}} L_{\mathbf{xy}}) + \eta (\delta - 1) \right) \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2. \end{aligned}$$

The minimum is achieved at

$$\eta = \frac{1 - \delta}{L_f + L_{\mathbf{xy}}\kappa_{\mathbf{xy}}}$$

Hence the decrease

$$V(\mathbf{x}^\eta, \mathbf{y}^\eta, \hat{\mathbf{x}}, \hat{\mathbf{y}}) \leq V(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}) - \frac{(1 - \delta)^2/2}{L_f + L_{\mathbf{xy}}\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2.$$

□

C.3. Proof of Lemma 4

Lemma 4. Assume that $\hat{\xi}_{k+1}$ satisfies Equations (14) and (15) with $\kappa_{\mathbf{xy}} > 0$ and that $\Delta_{k+1} > 0$. Then

$$\begin{aligned} V(\hat{\xi}_{k+1}, \xi_k) &\leq [r \gamma(\delta)(\kappa_f + \kappa_{\mathbf{xy}}^2) - 1] V(\xi_k, \tilde{\xi}_k), \\ \text{where } \gamma(\delta) &:= \frac{1 + (\delta/\kappa_{\mathbf{xy}})^2}{(1 - \delta)^2}. \end{aligned} \quad (20)$$

Proof. We recall Equation (15),

$$\left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k) \end{bmatrix} \right\| \leq \frac{\delta}{\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k) \end{bmatrix} \right\|. \quad (29)$$

Assume $\Delta_{k+1} \geq 0$, hence,

$$rV^{\text{Exact}}(\xi_k) \geq \frac{(1 - \delta)^2/2}{L_f + L_{\mathbf{xy}}\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k) \end{bmatrix} \right\|^2 \quad (30)$$

In this case, set $\xi_{k+1} = \hat{\xi}_{k+1}$ and $\tilde{\xi}_{k+1} = \xi_k$. Then

$$\begin{aligned} V(\xi_{k+1}, \tilde{\xi}_{k+1}) &= V(\hat{\xi}_{k+1}, \xi_k) \\ &= f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k) - f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1}) + \frac{1}{2\mu} \left(\|\nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1})\|^2 + \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k)\|^2 \right) \\ &\leq \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}_k) - \max_{\mathbf{y}} f(\mathbf{x}_k, \mathbf{y}) \\ &\quad + \frac{1}{2\mu} \left(\|\nabla_{\mathbf{x}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1})\|^2 \right) \\ &\quad + \frac{1}{2\mu} \left(\|\nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1})\|^2 + \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k)\|^2 \right) \\ &\stackrel{(29)}{\leq} \min_{\mathbf{x}} f(\mathbf{x}, \mathbf{y}_k) - \max_{\mathbf{y}} f(\mathbf{x}_k, \mathbf{y}) + \frac{1 + \left(\frac{\delta}{\kappa_{\mathbf{xy}}}\right)^2}{2\mu} \left(\|\nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1})\|^2 + \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k)\|^2 \right) \\ &\stackrel{\text{Def}}{=} -\min_{\hat{\xi}} V(\xi_k, \hat{\xi}) + \frac{1 + \left(\frac{\delta}{\kappa_{\mathbf{xy}}}\right)^2}{2\mu} \left(\|\nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1})\|^2 + \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k)\|^2 \right) \\ &\stackrel{(30)}{\leq} -\min_{\hat{\xi}} V(\xi_k, \hat{\xi}) + \frac{1 + \left(\frac{\delta}{\kappa_{\mathbf{xy}}}\right)^2}{2\mu} \frac{2(L_f + L_{\mathbf{xy}}\kappa_{\mathbf{xy}})r}{(1 - \delta)^2} \min_{\hat{\xi}} V(\xi_k, \hat{\xi}) \\ &\leq \left(\frac{1 + \left(\frac{\delta}{\kappa_{\mathbf{xy}}}\right)^2}{2\mu} \frac{2(L_f + L_{\mathbf{xy}}\kappa_{\mathbf{xy}})r}{(1 - \delta)^2} - 1 \right) V(\xi_k, \tilde{\xi}_k) \\ &= \left(\gamma(\delta)(\kappa_f + \kappa_{\mathbf{xy}}^2)r - 1 \right) V(\xi_k, \tilde{\xi}_k) \end{aligned}$$

where we assumed $\left(\gamma(\delta)(\kappa_f + \kappa_{\mathbf{xy}}^2)r - 1 \right) > 0$ in the last inequality.

□

C.4. Proof of Theorem 1

Theorem 1. Let f satisfy A. 1 and 2. After T iterations,

$$T \geq O\left(\kappa_f(\kappa_f + \kappa_{\mathbf{x}\mathbf{y}}^2) \log^2(\varepsilon^{-1})\right),$$

the output $\mathbf{x}_T, \mathbf{y}_T$ of Algorithm 1 with $\delta = \min\{\kappa_{\mathbf{x}\mathbf{y}}, \delta_{\max}\}$ guarantees that

$$V^{\text{exact}}(\mathbf{x}_T, \mathbf{y}_T) \leq V_T \leq \varepsilon.$$

Proof. From the subroutine in Algorithm 1, it is trivial that after the 1st or 2nd case (gradient step or swap), the Lyapunov contracts with a rate $1 - r$. It remains to bound the maximum number of iterations between two contractions, which we call T_{inner} .

Since we have seen that if $\hat{\xi}$ satisfies Equations (14) and (15), then either the gradient update in Equation (18) or the swapping technique described in Lemma 4 guarantees the contraction of the Lyapunov with factor $1 - r$, hence ending in either case 1 or case 2 in Algorithm 1.

Therefore, it remains to know how many iterations are required to compute a $\hat{\xi} = (\hat{\mathbf{x}}, \hat{\mathbf{y}})$ that achieves Equations (14) and (15), i.e.,

$$V(\xi_k, \hat{\xi}) - V^{\text{Exact}}(\xi_k) \leq (1 - r) \left(V(\xi_k, \tilde{\xi}_k) - V^{\text{Exact}}(\xi_k) \right), \quad (\text{Condition 1})$$

$$\left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\| \leq \frac{\delta}{\kappa_{\mathbf{x}\mathbf{y}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\|. \quad (\text{Condition 2})$$

In addition, because the algorithm is not terminated, we assume that $V(\xi_k, \tilde{\xi}) \geq \epsilon$, and therefore $V^{\text{Exact}}(\xi_k) \geq \epsilon$.

Complexity bound for condition 1: To simplify, let us call $\phi(\mathbf{z})$ a μ -PL and L -smooth function. Define the PL gap of gradient descent as

$$\phi^* = \min_{\mathbf{z}} \phi(\mathbf{z}), \quad 0 \leq G_t = \phi^* - \phi(\mathbf{z}_t) + \frac{\|\nabla \phi(\mathbf{z}_t)\|^2}{2\mu}, \quad \mathbf{z}_{t+1} = \mathbf{z}_t - \frac{1}{L} \nabla \phi(\mathbf{z}_t).$$

Since ϕ is smooth and PL,

$$\frac{1}{2L} \|\nabla \phi(\mathbf{z}_t)\|^2 \leq \phi(\mathbf{z}_t) - \phi^* \leq \frac{1}{2\mu} \|\nabla \phi(\mathbf{z}_t)\|^2.$$

Using the definition of G_t , the inequalities above, and Theorem 2,

$$G_t = \phi^* - \phi(\mathbf{z}_t) + \frac{\|\nabla \phi(\mathbf{z}_t)\|^2}{2\mu} \leq \left(\frac{L}{\mu} - 1 \right) (\phi(\mathbf{z}_t) - \phi^*) \leq \left(\frac{L}{\mu} - 1 \right) \left(1 - \frac{\mu}{L} \right)^t (\phi(\mathbf{z}_0) - \phi^*).$$

We can apply this result to $V(\xi_k, \hat{\xi})$, which gives

$$\begin{aligned} V(\xi_k, \hat{\xi}_t) - V^{\text{exact}}(\xi_k) &\leq \left(\frac{L_f}{\mu_f} - 1 \right) \left(1 - \frac{\mu_f}{L_f} \right)^{T_{\text{inner}}^{(1)}} \left(f(\tilde{\mathbf{x}}_k, \mathbf{y}_k) - \min_{\tilde{\mathbf{x}}} f(\tilde{\mathbf{x}}, \mathbf{y}_k) + \max_{\tilde{\mathbf{y}}} f(\mathbf{x}_k, \tilde{\mathbf{y}}) - f(\mathbf{x}_k, \tilde{\mathbf{y}}_k) \right) \\ &\leq \left(\frac{L_f}{\mu_f} - 1 \right) \left(1 - \frac{\mu_f}{L_f} \right)^{T_{\text{inner}}^{(1)}} \left(\frac{\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2}{2\mu_f} \right). \end{aligned}$$

Hence, we have to find $T_{\text{inner}}^{(1)}$ such that the following sufficient condition is satisfied,

$$\left(\frac{L_f}{\mu_f} - 1 \right) \left(1 - \frac{\mu_f}{L_f} \right)^{T_{\text{inner}}^{(1)}} \left(\frac{\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2}{2\mu_f} \right) \leq (1 - r) \left(V(\xi_k, \tilde{\xi}_k) - V^{\text{Exact}}(\xi_k) \right).$$

Writing the worst approximation ratio τ ,

$$\tau = \max_k \left\{ \frac{\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2}{V(\xi_k, \tilde{\xi}_k) - V^{\text{Exact}}(\xi_k)} \right\},$$

we obtain the following sufficient condition for $T_{\text{inner}}^{(1)}$,

$$T_{\text{inner}}^{(1)} \geq \kappa_f \log \left(\frac{\tau(\kappa_f - 1)}{2\mu_f(1-r)} \right) = O \left(\kappa_f \log \left(\frac{\kappa_f}{\epsilon} \right) \right),$$

where τ is uniformly bounded as demonstrated below.

Since the algorithm enforces a monotonic decrease,

$$V(\xi_k, \tilde{\xi}_k) \leq V(\xi_0, \tilde{\xi}_0),$$

and because

$$V(\xi_k, \tilde{\xi}_k) + V(\tilde{\xi}_k, \xi_k) = \frac{1}{2\mu_f} \left(\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2 \right) \quad (31)$$

$$+ \frac{1}{2\mu_f} \left(\|\nabla_{\mathbf{y}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{x}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2 \right) \quad (32)$$

we have

$$V(\xi_k, \tilde{\xi}_k) + V(\tilde{\xi}_k, \xi_k) \leq V(\xi_0, \tilde{\xi}_0) + V(\tilde{\xi}_0, \xi_0).$$

It follows that the gradient norms are uniformly bounded along the iterates:

$$\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2 \leq 2\mu_f [V(\xi_0, \tilde{\xi}_0) + V(\tilde{\xi}_0, \xi_0)] = O(V(\xi_0, \tilde{\xi}_0)). \quad (33)$$

The term $V(\xi_k, \tilde{\xi}_k) - V^{\text{Exact}}(\xi_k)$ is always lower bounded by ϵ for all iterations as the inner iterations will stop when the gap is less than ϵ . Thus, we have

$$\tau \leq O \left(\frac{V(\xi_0, \tilde{\xi}_0)}{\epsilon} \right).$$

Complexity bound for condition 2: It remains to know if the second condition can be met, i.e.,

$$\left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\| \leq \frac{\delta}{\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\|.$$

This condition seems harder as both sides depend on $\hat{\mathbf{x}}, \hat{\mathbf{y}}$. However, we will show that if the algorithm did not stop, then the right-hand side is lower bounded by a function of ϵ .

First, let us assume that the condition is not met, but $\hat{\mathbf{x}}, \hat{\mathbf{y}}$ achieves a small gradient, for instance,

$$\sqrt{\epsilon_{\nabla}} \geq \left\| \begin{bmatrix} \nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\| \geq \frac{\delta}{\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\|.$$

Let us look now at the sum of the two Lyapunov functions $V(\xi_k, \hat{\xi}_{k+1}) + V(\hat{\xi}_k, \xi_{k+1})$,

$$\begin{aligned} & V(\xi_k, \hat{\xi}_{k+1}) + V(\hat{\xi}_{k+1}, \xi_k), \\ &= \frac{1}{2\mu_f} \left(\|\nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1})\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}_{k+1})\|^2 + \|\nabla_{\mathbf{x}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}_{k+1}, \mathbf{y}_k)\|^2 \right), \\ &\leq \left(1 + \frac{\kappa_{\mathbf{xy}}^2}{\delta^2} \right) \frac{\epsilon_{\nabla}}{2\mu_f}. \end{aligned}$$

Therefore,

$$V(\xi_k, \hat{\xi}_{k+1}) \leq V(\xi_k, \hat{\xi}_{k+1}) + V(\hat{\xi}_{k+1}, \xi_k) \leq \left(1 + \frac{\kappa_{\mathbf{xy}}^2}{\delta^2} \right) \frac{\epsilon_{\nabla}}{2\mu_f}.$$

This means

$$\frac{\delta}{\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\| \leq \sqrt{\varepsilon \nabla}, \quad \varepsilon \nabla = \frac{2\mu_f}{1 + \frac{\kappa_{\mathbf{xy}}^2}{\delta^2}} \varepsilon, \quad \Rightarrow \quad V(\xi_k, \hat{\xi}_{k+1}) \leq \varepsilon. \quad (34)$$

and therefore will end the algorithm. This means in the worst case, we have to achieve,

$$\frac{\delta}{\kappa_{\mathbf{xy}}} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}_k, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}_k) \end{bmatrix} \right\| \leq \sqrt{\varepsilon \nabla}.$$

Using the rate on the gradient norm for gradient descent on smooth, PL functions (see Theorem 2 and Corollary 1), in the worst case,

$$\begin{aligned} T_{\text{inner}}^{(2)} &\geq \kappa_f \log \left(\kappa_f \frac{\delta^2}{\kappa_{\mathbf{xy}}^2} \frac{\|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2 + \|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2}{\varepsilon \nabla} \right), \\ &= \kappa_f \log \left(\kappa_f \frac{\delta^2}{\kappa_{\mathbf{xy}}^2} \left(1 + \frac{\kappa_{\mathbf{xy}}^2}{\delta^2} \right) \frac{\|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2 + \|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2}{2\mu_f \varepsilon} \right). \end{aligned}$$

We notice that, to balance both term, we can set

$$\delta = \min\{\delta_{\max}, \kappa_{\mathbf{xy}}\}, \quad 0 < \delta_{\max} < 1.$$

In this case, $\frac{\delta^2}{\kappa_{\mathbf{xy}}^2} < 1$ and the sufficient condition on $T_{\text{inner}}^{(2)}$ becomes

$$T_{\text{inner}}^{(2)} \geq \kappa_f \log \left(\kappa_f \frac{\|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2 + \|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2}{\mu_f \varepsilon} \right) = O(\kappa_f \log(\varepsilon^{-1})), \quad (35)$$

where $O(\kappa_f \log(\varepsilon^{-1}))$ follows from the fact that $\|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2 + \|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2$ is bounded above by $O(V(\xi_0, \tilde{\xi}_0))$ as shown in Equation (33).

Final complexity bound: This means, at worst, the algorithm has to run at most

$$T_{\text{inner}} = \max \left\{ T_{\text{inner}}^{(1)} + T_{\text{inner}}^{(2)} \right\} = O(\kappa_f \log(\varepsilon^{-1})).$$

iterations before guaranteeing a contraction in the Lyapunov.

Moreover, using Equations (12) and (22), we can derive the maximum number of Lyapunov contractions. From Equation (12), we have

$$V_{T_{\text{contr.}}} \leq (1 - r)^{T_{\text{contr.}}} V_0.$$

Now, to ensure $V_{T_{\text{contr.}}} \leq \varepsilon$, we have

$$T_{\text{contr.}} \geq \frac{\log \left(\frac{V_0}{\varepsilon} \right)}{-\log(1 - r)} \Rightarrow (1 - r)^{T_{\text{contr.}}} V_0 \leq \varepsilon.$$

We then derive a sufficient condition using the standard inequality

$$-\log(1 - r) \geq r \quad \text{for all } r \in (0, 1),$$

which yields the following more restrictive condition

$$T_{\text{contr.}} \geq \frac{\log \left(\frac{V_0}{\varepsilon} \right)}{r} \quad \left(\Rightarrow T_{\text{contr.}} \geq \frac{\log \left(\frac{V_0}{\varepsilon} \right)}{-\log(1 - r)} \right).$$

Thus, we have

$$T_{\text{contr.}} \geq \frac{1 + \gamma(\delta) (\kappa_f + \kappa_{\mathbf{xy}}^2)}{2} \log \left(\frac{V(\xi_0, \tilde{\xi}_0)}{\varepsilon} \right).$$

With $\delta = \min\{\delta_{\max}, \kappa_{\mathbf{xy}}\}$, we have that $\gamma(\delta)$ (see Equation (20)) can be bounded by

$$\gamma(\delta) = \frac{1 + (\delta/\kappa_{\mathbf{xy}})^2}{(1 - \delta)^2} \leq \frac{2}{(1 - \delta_{\max})^2}.$$

Therefore, we can obtain the following stronger condition on $T_{\text{contr.}}$ (because $\kappa_f > 1$),

$$T_{\text{contr.}} \geq (\kappa_f + \kappa_{\mathbf{xy}}^2) \frac{2}{(1 - \delta_{\max})^2} \log \left(\frac{V(\xi_0, \tilde{\xi}_0)}{\varepsilon} \right) = O\left((\kappa_f + \kappa_{\mathbf{xy}}^2) \log(\varepsilon^{-1})\right). \quad (36)$$

By multiplying Equation (35) with Equation (36), we have the following sufficient condition for the total number of iterations,

$$T_{\text{total}} \geq T_{\text{inner}} \cdot T_{\text{contr.}} = O\left(\kappa_f (\kappa_f + \kappa_{\mathbf{xy}}^2) \log^2(\varepsilon^{-1})\right).$$

□

Algorithm 2 SOTALDA: Second Order Tri-Action Lyapunov Descent Ascent

1: Main loop 2: Input: oracle of f, ϵ, $(\mathbf{x}_0, \mathbf{y}_0)$, μ_f, L_f, L_V, δ. 3: Init: $k \leftarrow 0$; $(\hat{\mathbf{x}}_0, \hat{\mathbf{y}}_0) \leftarrow (\mathbf{x}_0, \mathbf{y}_0)$. 4: $V_0 \leftarrow \frac{1}{2\mu} (\ \nabla_{\mathbf{x}} f(\mathbf{x}_0, \mathbf{y}_0)\ ^2 + \ \nabla_{\mathbf{y}} f(\mathbf{x}_0, \mathbf{y}_0)\ ^2)$. 5: while $V(\mathbf{x}_k, \mathbf{y}_k, \hat{\mathbf{x}}_k, \hat{\mathbf{y}}_k) \geq \epsilon$ do 6: $\hat{\mathbf{x}}_{k+1} \leftarrow \hat{\mathbf{x}}_k - \frac{1}{L_f} \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}_k, \mathbf{y}_k)$. 7: $\hat{\mathbf{y}}_{k+1} \leftarrow \hat{\mathbf{y}}_k + \frac{1}{L_f} \nabla_{\mathbf{y}} f(\mathbf{x}_k, \hat{\mathbf{y}}_k)$. 8: $(\mathbf{x}_{k+1}, \mathbf{y}_{k+1}, \hat{\mathbf{x}}_{k+1}, \hat{\mathbf{y}}_{k+1}, V_{k+1})$ 9: $\leftarrow \text{LyapUpdate}(\mathbf{x}_k, \mathbf{y}_k, \hat{\mathbf{x}}_{k+1}, \hat{\mathbf{y}}_{k+1}, V_k)$. 10: $k \leftarrow k + 1$. 11: end while 12: Output: $(\mathbf{x}_{\text{out}}, \mathbf{y}_{\text{out}}) \leftarrow (\mathbf{x}_k, \mathbf{y}_k)$.	1: Subroutine: LyapUpdate 2: Params: f, r from Eq. 40, $\eta = \frac{1}{L_V}$. 3: Input: $\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, V_{\text{old}}$. 4: Let $\mathbf{x}^{\text{grad}}$ be $\mathbf{x} - \eta \left(\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) + \frac{1}{\mu_f} \nabla_{\mathbf{xy}}^2 f(\mathbf{x}, \hat{\mathbf{y}}) \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \right).$ 5: Let $\mathbf{y}^{\text{grad}}$ be $\mathbf{y} + \eta \left(\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) - \frac{1}{\mu_f} \nabla_{\mathbf{yx}}^2 f(\hat{\mathbf{x}}, \mathbf{y}) \nabla_{\mathbf{x}} f(\hat{\mathbf{x}}, \mathbf{y}) \right)$ 6: $V_1 \leftarrow V(\mathbf{x}^{\text{grad}}, \mathbf{y}^{\text{grad}}, \hat{\mathbf{x}}, \hat{\mathbf{y}})$. 7: $V_2 \leftarrow V(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \mathbf{x}, \mathbf{y})$. 8: If $V_1 \leq \min\{V_2, (1-r)V_{\text{old}}\}$ (Grad is best) 9: Return $(\mathbf{x}^{\text{grad}}, \mathbf{y}^{\text{grad}}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, V_1)$ 10: If $V_2 \leq \min\{V_1, (1-r)V_{\text{old}}\}$ (Swap is best) 11: Return $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \mathbf{x}, \mathbf{y}, V_2)$ 12: Default: (Continue Approximation) 13: Return $(\mathbf{x}, \mathbf{y}, \hat{\mathbf{x}}, \hat{\mathbf{y}}, V_{\text{old}})$
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D. Algorithm design for smooth Lyapunov function

For completeness, we additionally assume that the Lyapunov function V defined in Equation (10) is smooth with constant L_V with respect to the main variables $(\mathbf{x}, \mathbf{y})$. Concretely, for all fixed $(\tilde{\mathbf{x}}, \tilde{\mathbf{y}})$, we assume

$$\begin{aligned}
 |V(\mathbf{x}_2, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) - V(\mathbf{x}_1, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) - \langle \nabla_{\mathbf{x}} V(\mathbf{x}_1, \mathbf{y}, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}), \mathbf{x}_2 - \mathbf{x}_1 \rangle| &\leq \frac{L_V}{2} \|\mathbf{x}_2 - \mathbf{x}_1\|^2, \\
 |V(\mathbf{x}, \mathbf{y}_2, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) - V(\mathbf{x}, \mathbf{y}_1, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}) - \langle \nabla_{\mathbf{y}} V(\mathbf{x}, \mathbf{y}_1, \tilde{\mathbf{x}}, \tilde{\mathbf{y}}), \mathbf{y}_2 - \mathbf{y}_1 \rangle| &\leq \frac{L_V}{2} \|\mathbf{y}_2 - \mathbf{y}_1\|^2.
 \end{aligned} \tag{37}$$

This assumption is strictly stronger than A. 1 and 3. Indeed, bounding the Lipschitz continuity of ∇V requires control over terms of the form

$$\left\| \nabla_{\mathbf{x}} f(\mathbf{x}_2, \hat{\mathbf{y}}) + \frac{1}{\mu_f} \nabla_{\mathbf{xy}}^2 f(\mathbf{x}_2, \hat{\mathbf{y}}) \nabla_{\mathbf{y}} f(\mathbf{x}_2, \hat{\mathbf{y}}) - \nabla_{\mathbf{x}} f(\mathbf{x}_1, \hat{\mathbf{y}}) + \frac{1}{\mu_f} \nabla_{\mathbf{xy}}^2 f(\mathbf{x}_1, \hat{\mathbf{y}}) \nabla_{\mathbf{y}} f(\mathbf{x}_1, \hat{\mathbf{y}}) \right\|$$

which explicitly involves the variation of the mixed Hessian $\nabla_{\mathbf{xy}}^2 f(\mathbf{x}_1, \mathbf{y}) - \nabla_{\mathbf{xy}}^2 f(\mathbf{x}_2, \mathbf{y})$. Such a bound cannot be guaranteed under first-order smoothness assumptions alone; the same difficulty arises symmetrically for the $\mathbf{y}$ variable.

One way to control this term is to assume that f has a Lipschitz-continuous Hessian, as is common in the analysis of second-order methods (Nesterov & Polyak, 2006; Nesterov, 2008). Since this additional regularity is not central to our main contributions, we instead adopt (37) directly as an assumption.

Under Equation (37) together with A. 3, the norm of the gradient-based update in Algorithm 2 satisfies

$$\begin{aligned}
 \|\mathbf{x}^{\text{grad}} - \mathbf{x}\| &= \frac{1}{L_V} \left\| \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) + \frac{1}{\mu_f} \nabla_{\mathbf{xy}}^2 f(\mathbf{x}, \hat{\mathbf{y}}) \nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}}) \right\| \\
 &\geq \frac{1}{L_V} \|\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})\| - \frac{1}{L_V \mu_f} \|\nabla_{\mathbf{xy}}^2 f(\mathbf{x}, \hat{\mathbf{y}})\| \|\nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}})\| \\
 &\geq \frac{1}{L_V} \|\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})\| - \frac{L_{\mathbf{xy}}}{L_V \mu_f} \|\nabla_{\mathbf{y}} f(\mathbf{x}, \hat{\mathbf{y}})\|.
 \end{aligned}$$

If $\hat{\xi}$ satisfies Equation (15), this yields

$$\|\mathbf{x}^{\text{grad}} - \mathbf{x}\| \geq \left(1 - \frac{\delta}{L_V}\right) \|\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})\|.$$

An analogous bound holds for the $\mathbf{y}$ update:

$$\|\mathbf{y}^{\text{grad}} - \mathbf{y}\| \geq \left(1 - \frac{\delta}{L_V}\right) \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y})\|.$$

Applying the descent lemma to V then gives

$$V(\xi^{\text{grad}}, \hat{\xi}) - V(\xi, \hat{\xi}) \leq -\frac{\|\xi^{\text{grad}} - \xi\|^2}{2} \leq -\left(1 - \frac{\delta}{L_V}\right)^2 \left(\|\nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}})\|^2 + \|\nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y})\|^2\right),$$

which matches the condition in Lemma 3 with L_V replacing $L_f + L_{\mathbf{xy}}\kappa_{\mathbf{xy}}$.

Accordingly, the quantity Δ_{k+1} in Equation (17) becomes

$$\Delta_{k+1} = rV^{\text{Exact}}(\xi_k) - \frac{(1-\delta)^2/2}{L_V} \left\| \begin{bmatrix} \nabla_{\mathbf{x}} f(\mathbf{x}, \hat{\mathbf{y}}) \\ \nabla_{\mathbf{y}} f(\hat{\mathbf{x}}, \mathbf{y}) \end{bmatrix} \right\|^2. \quad (38)$$

As a consequence, using the proof technique from Lemma 4, we have

$$V(\hat{\xi}_{k+1}, \xi_k) \leq [r\kappa_V\gamma(\delta) - 1]V(\xi_k, \hat{\xi}_k), \quad \kappa_V := \frac{L_V}{\mu_f}, \quad \gamma(\delta) = \frac{1 + (\delta/\kappa_{\mathbf{xy}})^2}{(1-\delta)^2}. \quad (39)$$

Balancing the two regimes as in Equations (19) and (38), the convergence remains linear with rate

$$r_V = \frac{2}{1 + \gamma(\delta)\kappa_V}. \quad (40)$$

Following the same reasoning as in Theorem 1, after $T \geq O(\kappa_f\kappa_V \log(\varepsilon^{-1}))$ iterations, we obtain $V_T \leq \varepsilon$.

E. Other Algorithms

Algorithm 3 Direct minimization of V using gradient descent (conceptual).

Input: Initial point $(\mathbf{x}_0, \mathbf{y}_0) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_y}$, $\mathbf{x} = \tilde{\mathbf{x}} = \mathbf{x}_0$ and $\mathbf{y} = \tilde{\mathbf{y}} = \mathbf{y}_0$, PL constant μ , Stepsize $\eta_{\mathbf{x}}, \eta_{\mathbf{y}}, \eta_{\tilde{\mathbf{x}}}, \eta_{\tilde{\mathbf{y}}}$,

Initialize: $k \leftarrow 0$

while $V(\mathbf{x}_k, \mathbf{y}_k, \tilde{\mathbf{x}}_k, \tilde{\mathbf{y}}_k) = f(\mathbf{x}_k, \tilde{\mathbf{y}}_k) - f(\tilde{\mathbf{x}}_k, \mathbf{y}_k) + \frac{1}{2\mu} (\|\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k)\|^2 + \|\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k)\|^2) > \epsilon$ **do**

$$\tilde{\mathbf{x}}_{k+1} \leftarrow \tilde{\mathbf{x}}_k - \eta_{\tilde{\mathbf{x}}} \left[-\nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k) + \frac{1}{\mu} \nabla_{\mathbf{x}\mathbf{x}}^2 f(\tilde{\mathbf{x}}_k, \mathbf{y}_k) \nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_k, \mathbf{y}_k) \right]$$

$$\tilde{\mathbf{y}}_{k+1} \leftarrow \tilde{\mathbf{y}}_k - \eta_{\tilde{\mathbf{y}}} \left[\nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k) + \frac{1}{\mu} \nabla_{\mathbf{y}\mathbf{y}}^2 f(\mathbf{x}_k, \tilde{\mathbf{y}}_k) \nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_k) \right]$$

$$\mathbf{x}_{k+1} \leftarrow \mathbf{x}_k - \eta_{\mathbf{x}} \left[\nabla_{\mathbf{x}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_{k+1}) + \frac{1}{\mu} \nabla_{\mathbf{x}\mathbf{y}}^2 f(\mathbf{x}_k, \tilde{\mathbf{y}}_{k+1}) \nabla_{\mathbf{y}} f(\mathbf{x}_k, \tilde{\mathbf{y}}_{k+1}) \right]$$

$$\mathbf{y}_{k+1} \leftarrow \mathbf{y}_k - \eta_{\mathbf{y}} \left[-\nabla_{\mathbf{y}} f(\tilde{\mathbf{x}}_{k+1}, \mathbf{y}_k) + \frac{1}{\mu} \nabla_{\mathbf{y}\mathbf{x}}^2 f(\tilde{\mathbf{x}}_{k+1}, \mathbf{y}_k) \nabla_{\mathbf{x}} f(\tilde{\mathbf{x}}_{k+1}, \mathbf{y}_k) \right]$$

$$k \leftarrow k + 1$$

end while

Output: $\mathbf{x}_k, \mathbf{y}_k$

Algorithm 4 First-order approximation of a Hessian-vector product

Input: First-order oracle ∇g (smooth with constant L_g), point $\mathbf{x}$, vector $\mathbf{v}$, error tolerance ϵ

Set step size $h \leftarrow \frac{2\epsilon}{L_g \|\mathbf{v}\|^2}$

Compute shifted point $\mathbf{x}_+ \leftarrow \mathbf{x} + h\mathbf{v}$

Estimate product $\mathbf{w} \leftarrow \frac{\nabla g(\mathbf{x}_+) - \nabla g(\mathbf{x})}{h}$

Output: Approximation $\mathbf{w} \approx \nabla^2 g(\mathbf{x})\mathbf{v}$ such that $\|\nabla^2 g(\mathbf{x})\mathbf{v} - \mathbf{w}\| \leq \epsilon$ (see Lemma 7)
